Commercial Airplane Single Engine Land (CASEL) 141.55e

04/07/2025 (Rev 8)

Chapter 1 – Commercial 55e Single Engine Training Course Outline (TCO)

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Record of Revisions

REV NO.	DATE	INITIALS	LIST of CHANGES
ISSUE	03/13/2012	SP	
REV 1	08/03/2012	SP	
REV 2	04/02/2013	SP	
REV 3	11/04/2015	JL	
REV 4	05/17/2017	NS	
REV 5	04/20/2018	NS	
REV 6	12/20/2020	CG	Pg. 4- Changed format to record of revisions Pg. 1-30- Entire doc name change/ logo to “L3Harris” Pg. 5- removed reference to preamble Pg. 6- removed SR22, modified PA28R to PA28; added ref. to 141.39 for AC used; removed Mode C and Added ADSB to transponder, removed Autopilot requ Pg. 7,14 & 25- removed Lambert, Dugger and Gyuris added Crutchfield, respectively. Pg 8 – replaced Hurley with Hilmer as Chief Ground, added Thompson as Asst. Chief Ground Pg. 10 amended Course Overview “Training Syllabus”, Changed ref. to stages 2-Ground and 2-flight (previous ref. was error); added NOTE for JU students Pg. 13- removed Computer Based Training (CBT) from Ground Training; added “Training Manager” as designee for lesson sequence. Change; Pg. 18-22 and 28- Updated Facility Diagrams; Pg. 24 added Frasca RTD Pg. 25- Updated Jacksonville Base Address (same facility address changed) Pg. 27- removed Aerosim AATD added Frasca RTD
REV 7	11/04/2024	DK	Changed all Chief and Assitant Chief Flight Instructor references to refer to the current Webopss A006 an A007.
REV 8	04/07/2025	DK	Changed all references from L3Harris to Acron

After revision insertion, enter the date the revision is incorporated and initial in the appropriate column.

FAA APPROVED

PAULINE AYN O'CONNOR

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ORLANDO FSDO-SO15

List of Effective Pages

CAMEL 55e Chapter 1					
PAGE	REVISION	DATE	PAGE	REVISION	DATE
1	8	04/07/2025	2	8	04/07/2025
3	8	04/07/2025	4	8	04/07/2025
5	8	04/07/2025	6	8	04/07/2025
7	8	04/07/2025	8	8	04/07/2025
9	8	04/07/2025	10	8	04/07/2025
11	8	04/07/2025	12	8	04/07/2025
13	8	04/07/2025	14	8	04/07/2025
15	8	04/07/2025	16	8	04/07/2025
18	8	04/07/2025	17	8	04/07/2025
19	8	04/07/2025	20	8	04/07/2025
21	8	04/07/2025	22	8	04/07/2025
23	8	04/07/2025	24	8	04/07/2025
25	8	04/07/2025	26	8	04/07/2025
27	8	04/07/2025	28	8	04/07/2025
29	8	04/07/2025	30	8	04/07/2025

Revision System

Commercial Airplane Single Engine Syllabus revisions will be issued in consecutive numbers; and each page of the new revision will contain the revision number and the date the revision was issued.

It is the responsibility of all affected personnel to periodically review the *Commercial Airplane Single Engine Syllabus* to ensure that all current procedures are being followed.

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ORLANDO FSDO-SO15

Aircraft

The aircraft that may be used in this course are:

Cessna 172 (4 place)
Cirrus SR-20 (4 place)
Piper PA-28 (4 place)

All aircraft used in this course comply with FAR 141.39 and must:

1. Be registered as a civil aircraft of the United States and certified in the standard airworthiness category.
2. Be maintained and inspected in accordance with the requirements of 14 CFR Part 91.
3. Have engine power controls and flight controls easily accessible and that operate in a normal manner from each pilot station.

Required Aircraft Equipment

1. Transponder with Mode S
2. Equipped with ADS-B IN and OUT or OUT
3. MFD and PFD or standard instrument package
4. GPS or ADF

Personnel

Chief Flight Instructor

The Chief Flight Instructor designated for this course is found in the current Webopss A006 and A007. The Chief Flight Instructor meets the requirements outlined in 14 CFR 141.35.

Assistant Chief Flight Instructors

The Assistant Chief Flight Instructor(s) designated for this course are found in the current Webopss A006 and A007.

The Assistant Chief Flight Instructor(s) designated for this course must meet the requirements outlined in 14 CFR 141.36 and will be designated by letter.

Flight Instructors

Each flight instructor assigned to this course must have completed Flight Instructor Training and be the holder of at least a Commercial Pilot Certificate with an Airplane category and Single Engine Land class rating, as appropriate. In addition, each instructor must hold a Flight Instructor Certificate with an Airplane category and Single Engine class rating, as appropriate, with an Instrument Rating.

Chief Ground Instructor

The Chief Ground Instructor for this course is found in the current Webopss A006 and A007. The Chief Ground Instructor meets the requirements outlined in 14 CFR 141.35(e).

Whenever a Chief or Assistant Chief Ground Instructor is either undesignated or unavailable, the appropriate Chief or Assistant Chief Flight Instructor may assume those duties.

Assistant Chief Ground Instructors

The Assistant Chief Ground Instructor(s) designated for this course is found in the current Webopss A006 and A007.

The designated Assistant Chief Ground Instructor(s) for this course meets the requirements outlined in 14 CFR 141.36(e) and will be designated by letter.

Student Information

Course Enrollment

To be enrolled in this course, a crewmember must hold a current private pilot certificate with instrument rating and at least a third class medical certificate prior to beginning the flight training portion of the course.

Requirements for Graduation

Crewmembers must satisfactorily complete the training outlined in this syllabus and be at least 18 years of age.

Lesson Description and Stages of Training

Each lesson is fully described within the syllabus, including the objectives, standards, and measurable units of accomplishment and learning. The stage objectives and standards are described at the beginning of each stage within the syllabus.

Tests and Checks

The syllabus incorporates stage checks and End of Course Tests in accordance with 14 CFR Part 141. These checks are given by the Chief Instructor, a designated Assistant Chief, or another designated Check Instructor as appropriate.

Course Overview

Course Objective

The student will obtain the knowledge, skill, and aeronautical experience necessary to meet the requirements for a Commercial Pilot Airplane Single Engine Land.

Course Completion Standards

The student will demonstrate through written, oral, and flight tests that the knowledge, skill, and experience requirements necessary to obtain a Commercial Pilot Airplane are met.

Training Syllabus

The syllabus used will be the most current revision of the Acron Aviation Academy Inc. FAA Approved syllabus. (Chapters 2 and 3 of this TCO)

The training syllabus meets all curriculum requirements as prescribed by 14 CFR Part 141, Appendix D. It is composed of separate ground and flight courses to allow greater flexibility and to accommodate individual needs. The training courses may be taught either separately or concurrently.

Training Course

The Ground Training Course contains two (2) stages. The Flight Training Course contains two (2) stages.

NOTE

Students enrolled at Jacksonville University and completing a Bachelor of Science Degree in Aviation Management and Flight Operations receive full credit for all Ground School requirements for the following courses: Private ASEL, Instrument Airplane, Commercial Multi-Engine, CFI-A, and CFI-I.

Course Introduction

The Commercial ASEL Course utilizes the building-block theory of learning, which recognizes that each item taught must be presented on the basis of previously learned knowledge and skills.

Course Elements

The Commercial ASEL Course utilizes separate flight and ground segments. The course includes the latest FAA instructor certification requirements and a maximum of student-oriented instruction. The syllabus and support materials not only provide necessary information, but also guide the student through the course in a logical manner.

Airplane Practice

Airplane practice must be conducted so that the student obtains the maximum benefit from each flight. Prior to each solo flight, an instructor will carefully instruct the student in the maneuvers to be performed during the flight and what is to be accomplished. This guidance will ensure that the student receives maximum benefit from the solo flight.

Use of Flight Training Devices

The Commercial ASEL Course allows, as an option, the use of approved flight training devices for those lessons designated in the Course Time Allocation Table under the Aircraft Type heading. The best results are obtained when a flight training device introduction is accomplished just prior to the flight lesson.

Preflight Briefing

The preflight briefing is an integral part of the learning process. During the preflight briefing, the PUI (Pilot under training) firstly brief the IP (Instructor Pilot). During this briefing the PUI will brief the IP on the possible hazards to the route of flight, review current weather, current personal minimums and any other items of importance. Once the PUI has conducted their briefing the IP must review the items and challenge the information with specific questions. Once this is complete the IP will thoroughly brief the PUI on the scenario, and all of the listed tasks and completion requirements. The "*Preflight Briefing*" should take a minimum of 15 minutes per flight/flight training device event.

Postflight Evaluation

The postflight evaluation is equally as important as the preflight briefing. Each scenario must be thoroughly debriefed, during which time the PUI will evaluate their performance and debrief the IP. The importance of this "*Self Evaluation*" is to help the PUI learn to be critically honest of their own performance and to learn to come up with possible ways to correct any deficiencies without input from the IP. The IP will then debrief the PUI while emphasizing areas where the IP saw deficiencies, or where there was a notable difference in the "*Self Evaluation*" by the PUI and the evaluation of the IP. Noticeable advancement should be apparent and recommendations should be made for improvement, where appropriate. This action is a valuable instructional technique because it increases retention and helps to prepare the student for the next lesson. The "*Postflight Evaluation*" should take a minimum of 15 minutes per flight/flight training device event.

Student Stage Checks

Stage checks measure the student's accomplishments during each stage of training. The conduct of each stage check is the responsibility of the Chief Instructor. The authority to conduct these tests may be delegated to an appropriately designated and approved Assistant Chief Instructor or Check Instructor. This procedure provides close supervision of training and another opinion on the student's progress. The stage check also gives the Chief Instructor an opportunity to check the effectiveness of the instructors and their teaching methods.

An examination of the building-block theory of learning will show that it is extremely important for progress and proficiency to be satisfactory before the student enters a new stage of training.
Therefore, the next stage should not begin until the student successfully completes the current stage. Failure to follow this progression may defeat the purpose of the stage check and lead to overall course breakdown.

Ground Training

In accordance with 14 CFR Part 141, ground training is an integral part of instructor certification courses. The ground training syllabus in this course has been designed to meet this requirement.

The Ground Training Course may be presented to the student either as a formal classroom program or as an individual program. This allows for greater flexibility and accommodation to individual student needs.

Using the Ground Lessons

The ground lessons are generally divided into three sections--Lesson Introduction, Lecture, and Class Discussion. During the introduction, the instructor should outline the subject material to be covered during the training session, the objective for learning that information, and the performance standards necessary for successful lesson completion.

Textbooks

Prior to each ground lesson, the student should read and study the appropriate textbook section or chapter, as assigned. The appropriate textbook section or chapter will contain the information necessary to help the student complete the academic stages of the Commercial ASEL Course.

DVD Presentations

Following the ground lesson presentations, the student, as an option, may view DVD presentations. After the DVD presentation, the student should feel free to ask questions and obtain clarification of subject material viewed.

Final Examination

When all appropriate ground lesson presentations are complete, the student must take the Training Course Final Exam. Following the test, the instructor should assign the student review of appropriate subject areas. After a thorough review, the actual FAA Knowledge Test, if applicable, should be taken.

CASEL 55e- Training Course Implementation

The lesson sequence and content have been designed to provide the student with maximum training prior to the introduction of new maneuvers or procedures.

If any deviation from the normal lesson progression is desired, only the Chief Instructor, designated Assistant Chief or Training Manager for the course may grant approval.

NOTE

If the student is completing the ground training and flight training courses concurrently, the instructor must ensure that the student's flight training does not progress faster than the ground training.

Sanford Facility (SFB)

Acron Aviation Academy's Sanford base is located at the Orlando Sanford International Airport, 2700 Flightline Avenue, Sanford, FL 32773.

Chief and Designated Assistant Chief Flight Instructor

The Chief Flight Instructor for Acron Aviation Academy is found in Webopss A006 and A007.

Airport

The Orlando Sanford International Airport (SFB) is the main operations base for training. The airport has four (4) hard surface runways and meets the requirements of 14 CFR Part 141.38 for day and night flight operations.

A federally operated control tower operates daily between 0630-2300 hours. The airport has ILS and GPS approaches.

Fuel and maintenance services are available daily between 0600-2300 hours.

Training Airports

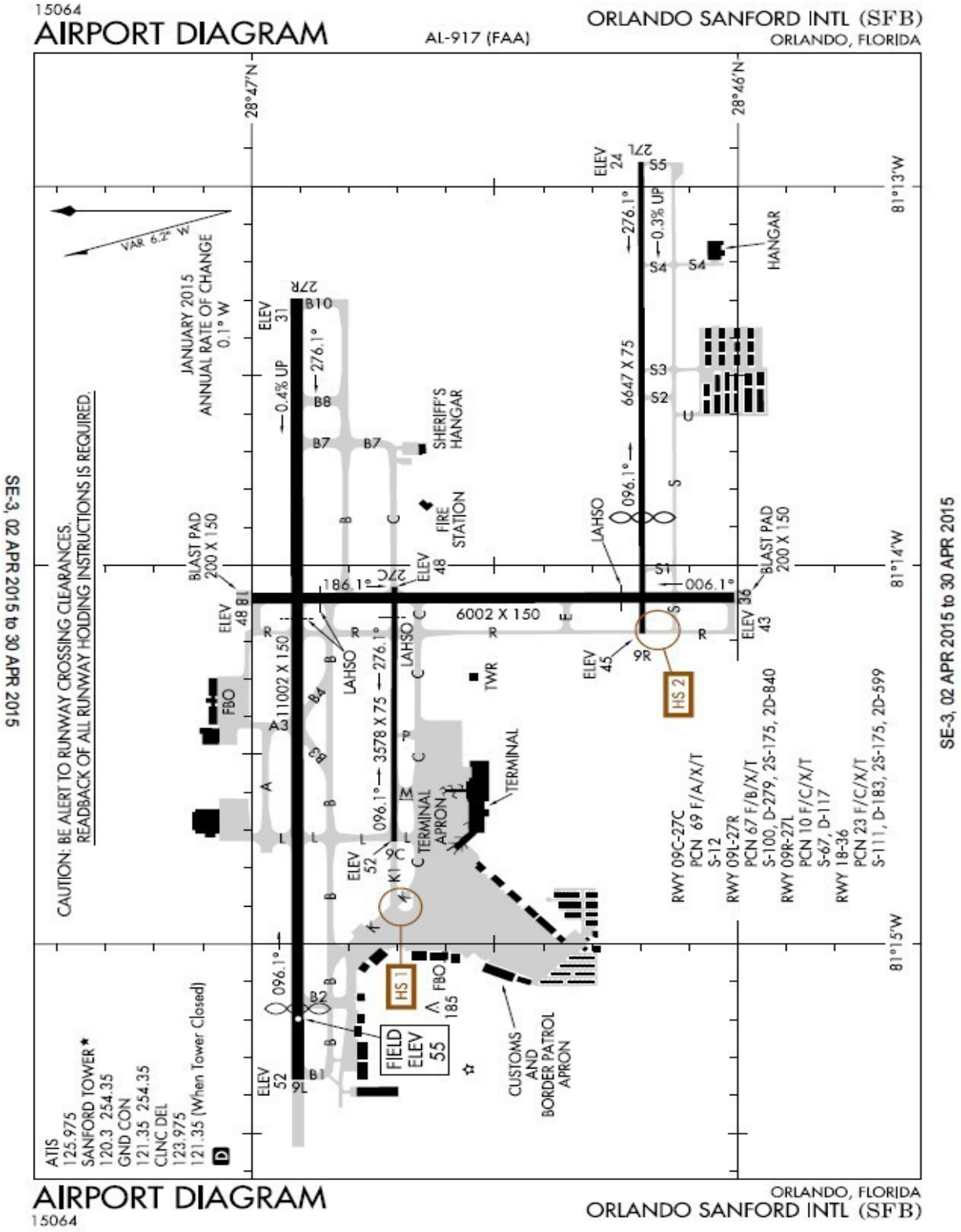
All training airports meet the requirements outlined in 14 CFR Part 141.38.

Direction is given to the Academy's flight instructors through an Approved Airports list in supplemental documentation.

The Chief Flight Instructor or designated Assistant Chief Flight Instructor may approve any appropriate public use airport listed in the Chart Supplement.

Sanford Airport Diagram

NOTE Diagram not current and not to be used for navigation.



Operations Center

Acron Aviation Academy's Operations Center is the focal point for all flight lesson activity. Adequate areas have been provided for preflight briefings, dispatch, and postflight briefings of all flight lesson activities.

The Operations Center includes a Planning Area and Briefing Rooms. The Flight/Ground Briefing Center provides additional space for briefings and small group instruction. Each briefing area is equipped with an erasable marker board as well as tables and chairs to accommodate students and instructors.

Planning Area
This area is equipped with tables and chairs adequate for both students and instructors.

Telephones are available to provide communication to the nearest FAA Flight Service Station. In addition, weather stations are also available for preflight planning purposes.

Sanford Simulator Training Building

Acron Aviation Academy's Simulator Training Building is the focal point for all simulator and flight training device activity. Adequate areas have been provided for AATD and FTD training lessons.

The Simulator Training Building includes Briefing Rooms. Each briefing area is equipped with an erasable marker board as well as tables and chairs to accommodate students and instructors.

NOTE

All rooms and areas conform to local building, sanitation, and health codes. All rooms are designed and located so that students will not be distracted by the instruction conducted in other rooms or by other flight and maintenance operations on the airport.

Academics

The Academics facilities are located in the Academics Building. This building houses classrooms, and the Learning Resource System.

Classrooms

Classrooms are equipped with tables and chairs to accommodate students. Each classroom contains at least one (1) chalkboard or erasable marker board of adequate size and one (1) projection screen.

Learning Resource Center

The Learning Resource Center (LRC) is equipped with tables and chairs to accommodate students. The LRC is available to students wishing to accomplish additional studying in a group or with a tutor. In addition, the LRS provides computer stations to students wishing to accomplish study using CBT materials, or by the viewing of DVD's.

Computer Testing Building

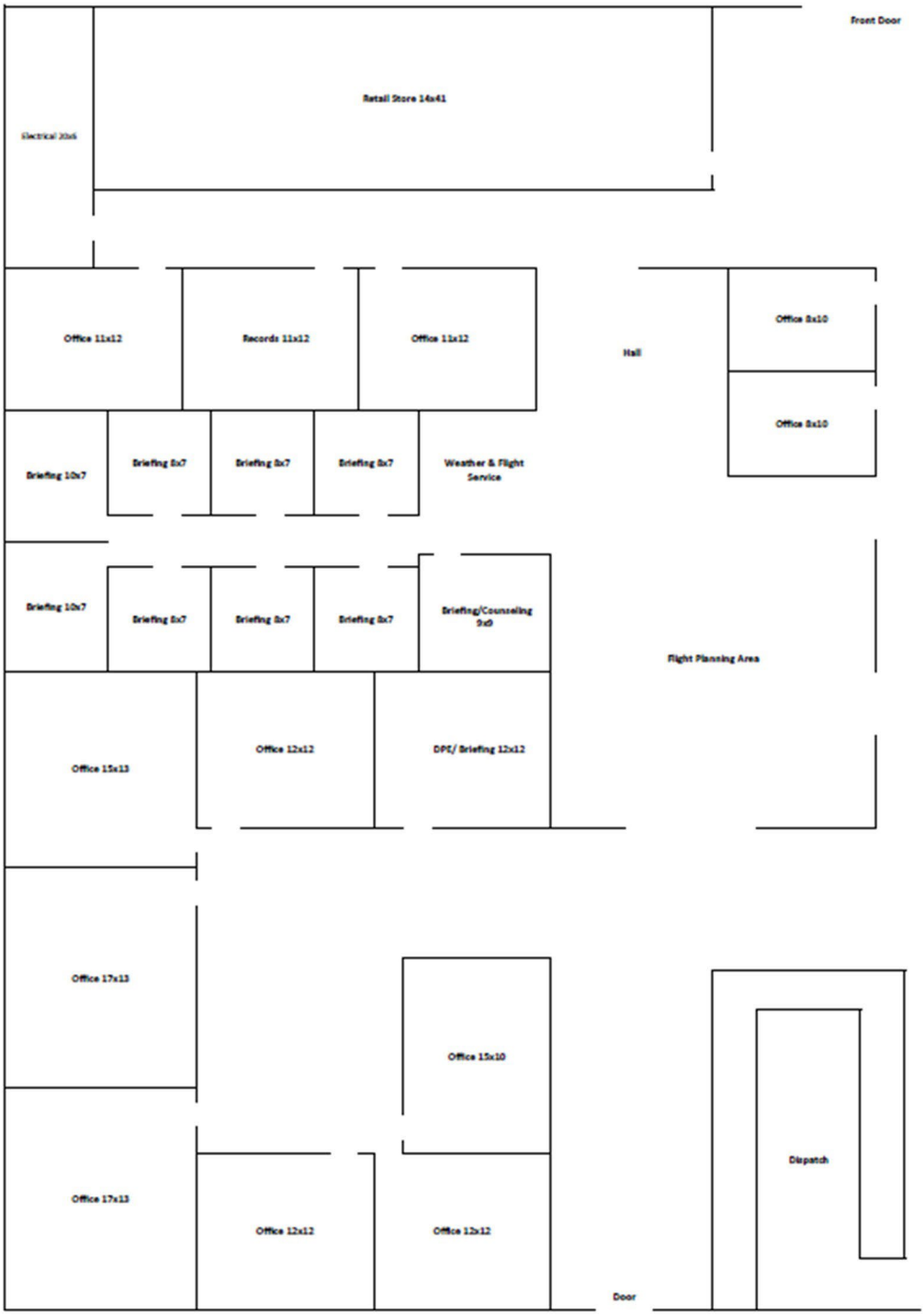
The Computer Testing Building is located at Bldg 41 2749 Flightline Ave, Sanford, Fl 32773. The Computer Testing Building has been set aside for the purpose of administering computerized FAA Knowledge Test Examinations.

NOTE

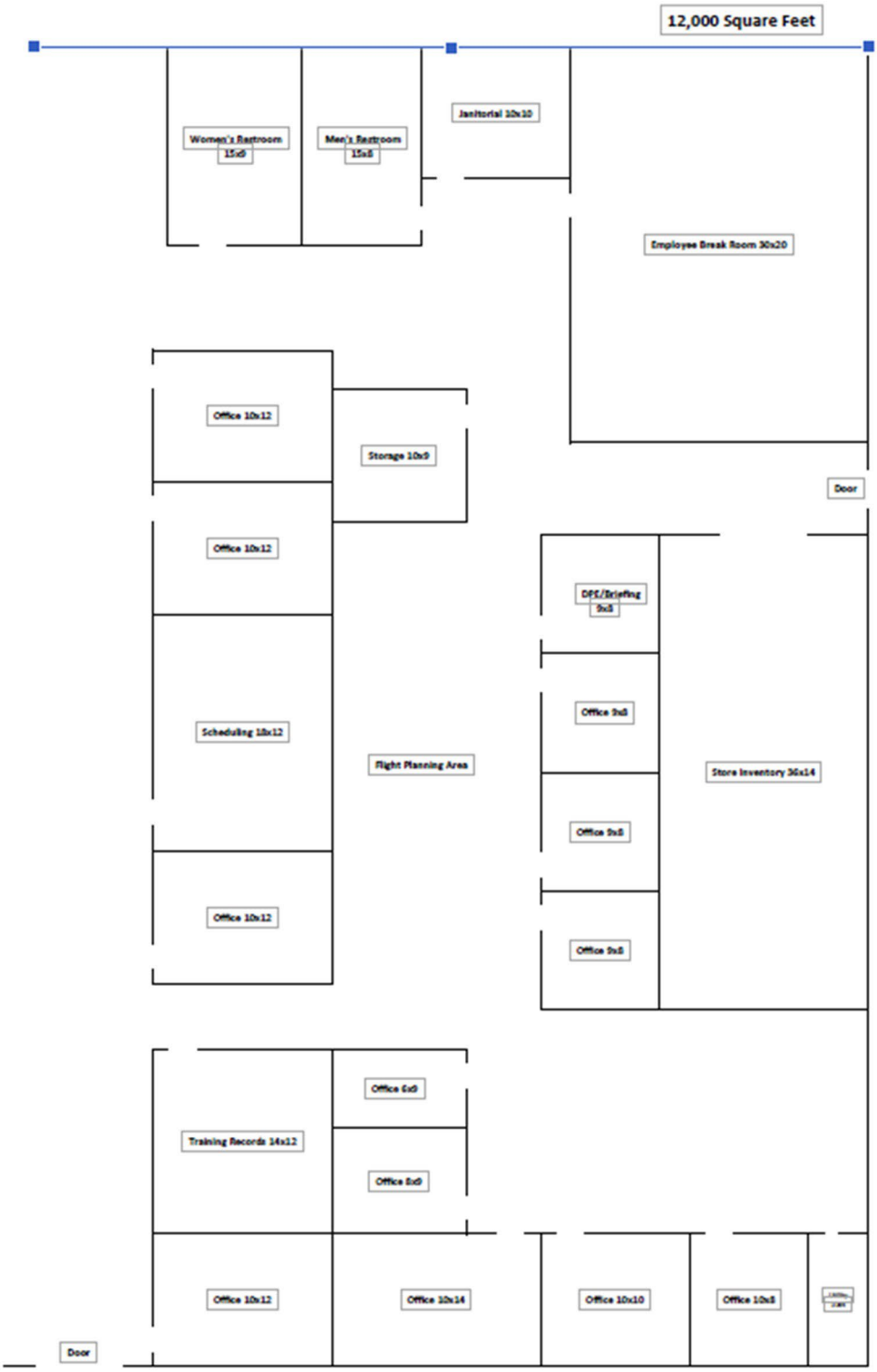
All rooms and areas conform to local building, sanitation, and health codes. All rooms are designed and located so that students will not be distracted by the instruction conducted in other rooms or by other flight and maintenance operations on the airport.

Building Diagrams

Sanford Operations Building – East Section

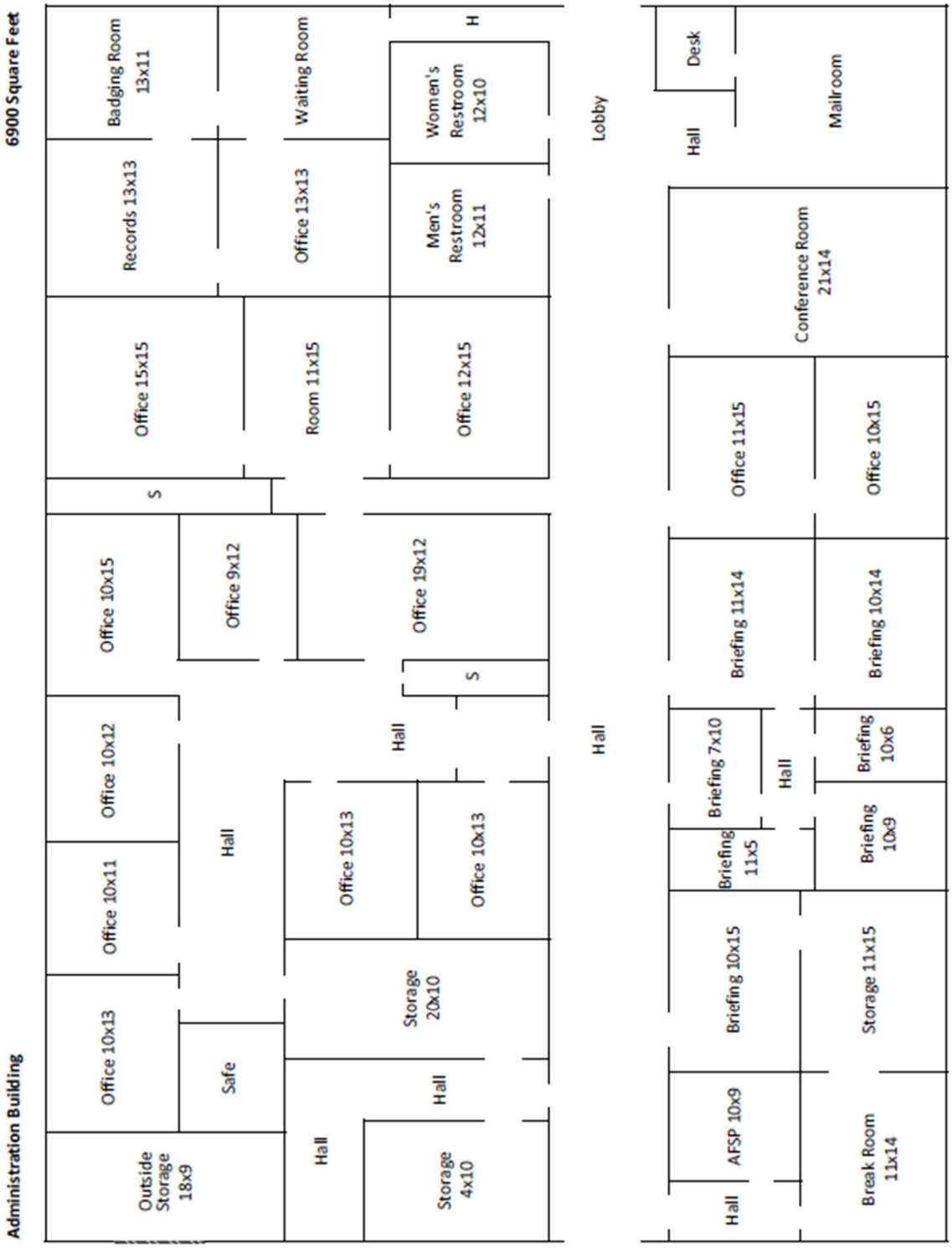


Sanford Operations Building – West Section

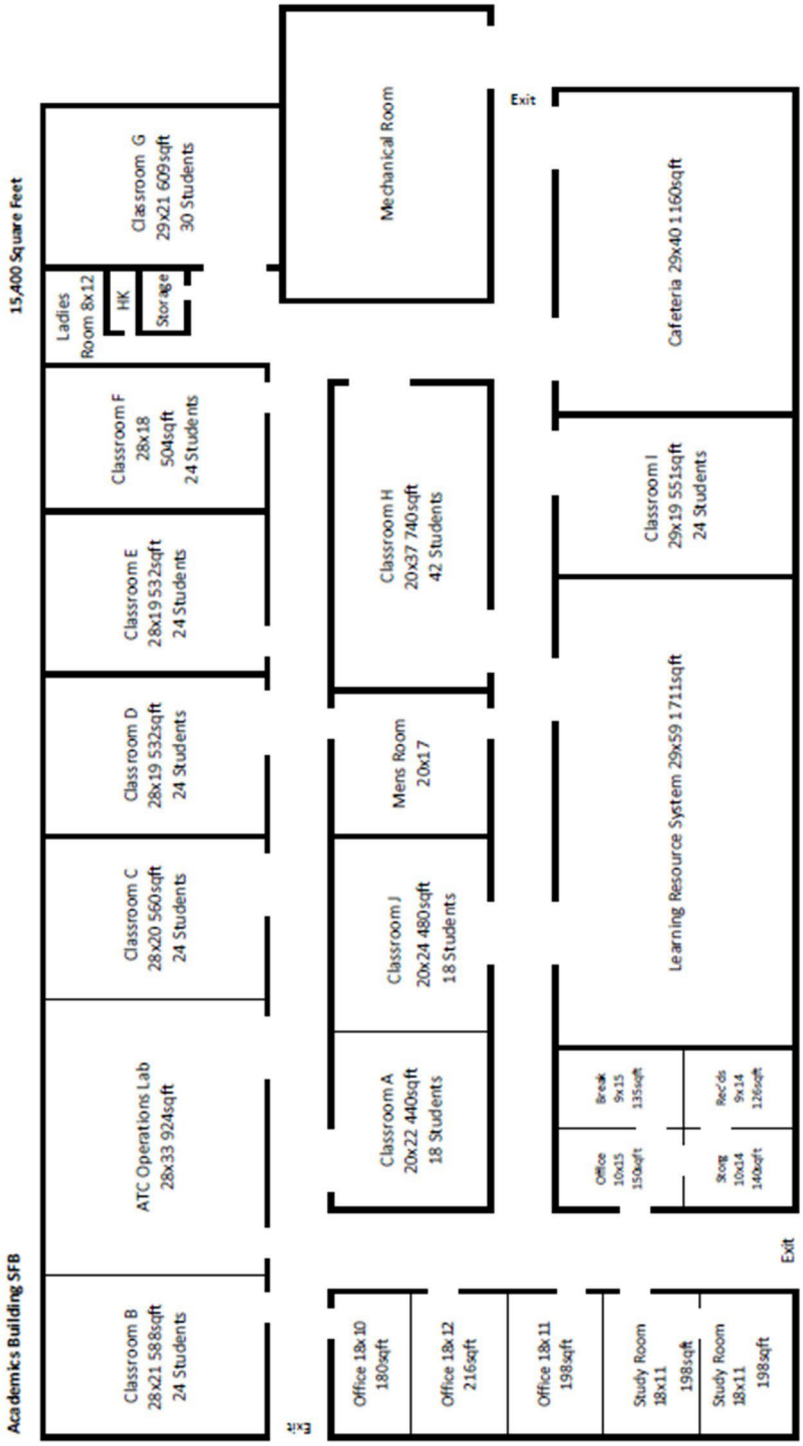


Sanford Administration Building

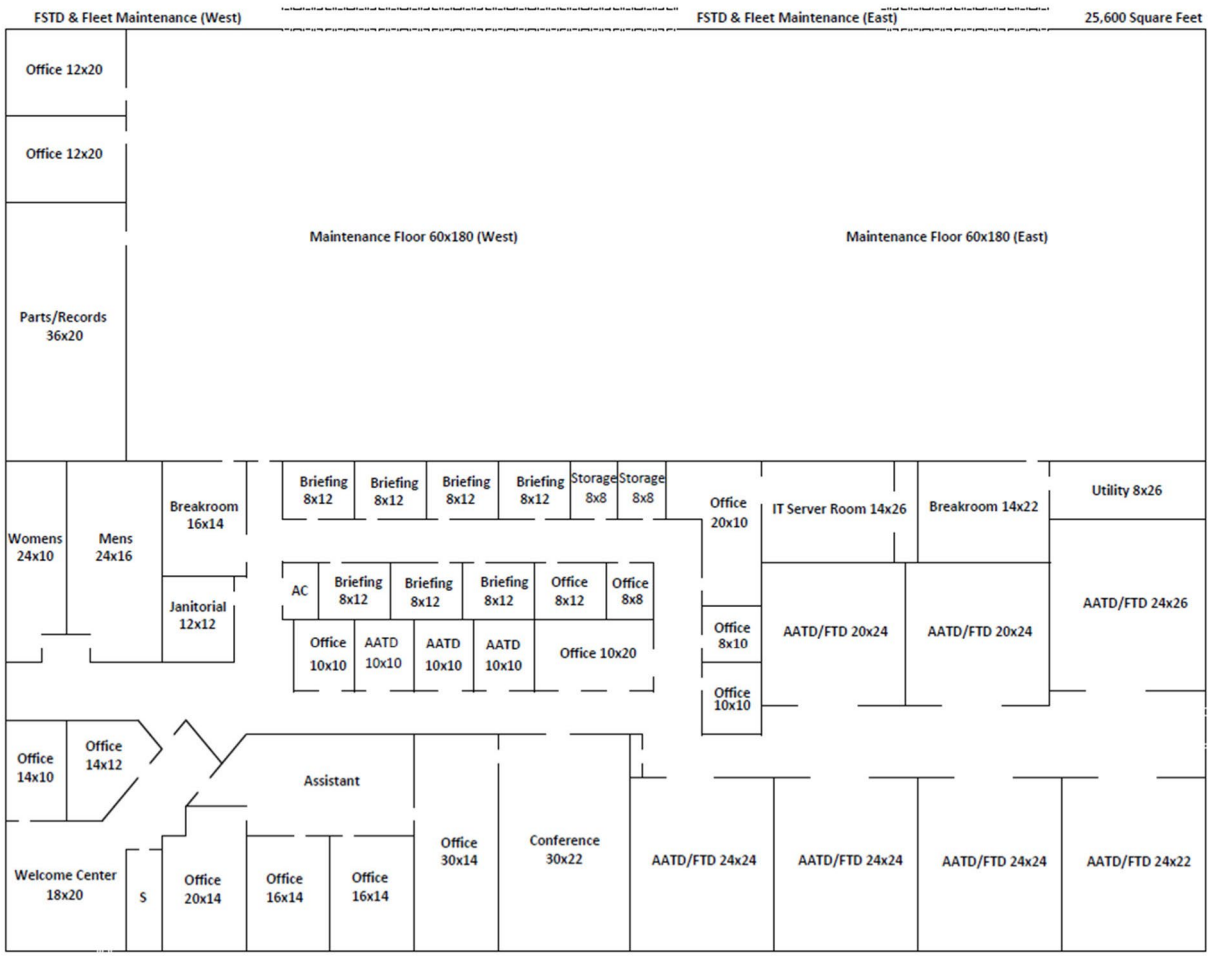
NOTE building not currently in use for academy operations, refserved for future use.



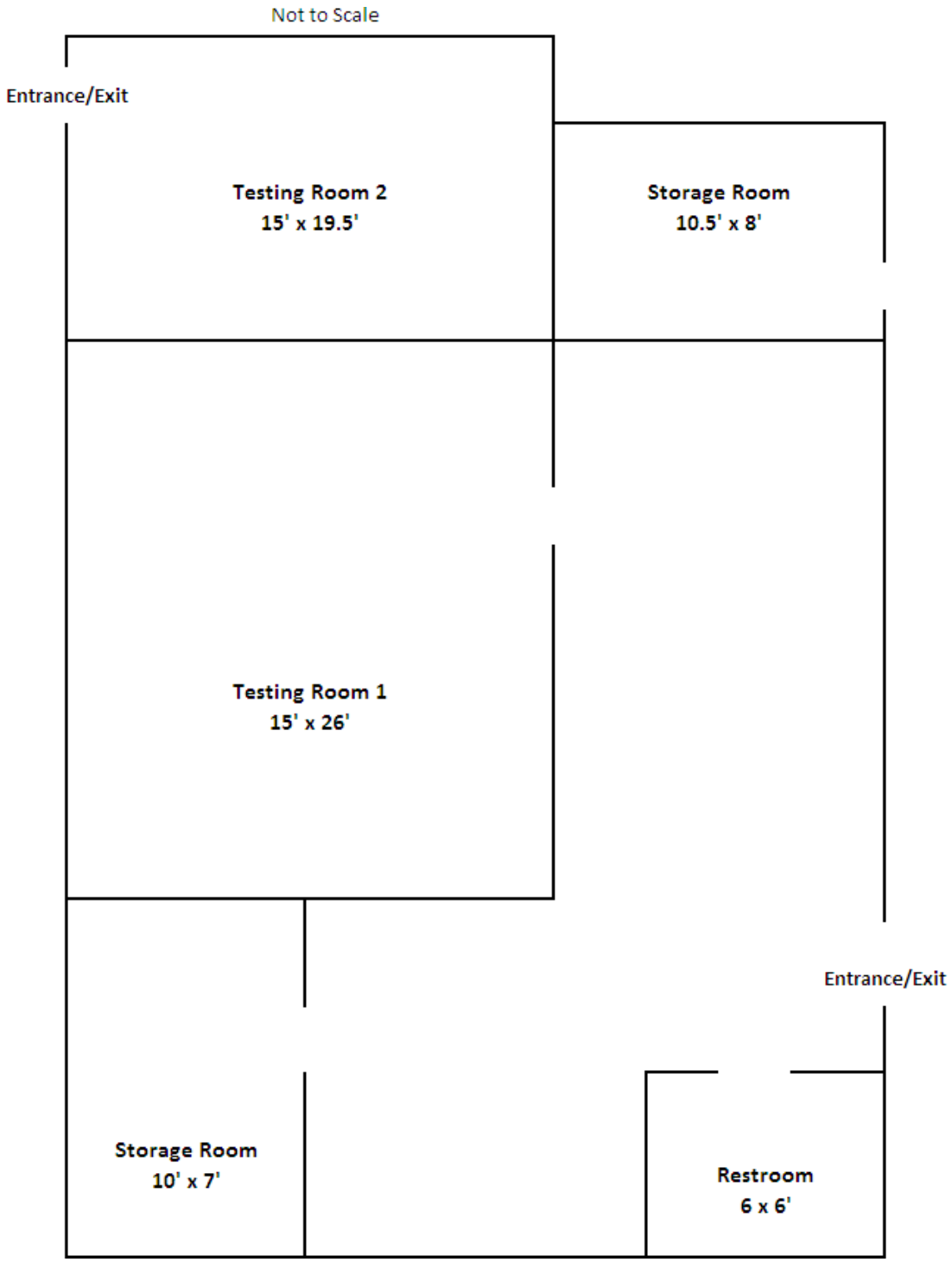
Sanford Academics Building



Sanford Simulator Training Building



Sanford Computer Testing Building



Ground Training Aids

The training aids available for use, as appropriate, are as follows:

A. Audiovisual Aids
   Computers with projectors, overhead projectors, Interactive Smart Boards, Diagrams and Charts
B. Mockups
   Large, presentation-type, E6B flight computer and Cockpit Procedures Trainer
C. Aircraft Components
   Flight instruments and aircraft engines on stands

In addition, for enhancement and reinforcement of training a variety of video tapes and computer- based training is available

Flight Training Devices

The Academy's flight training devices/approved aviation training devices meet the requirements specified in 14 CFR Part 141.41 (b)/14 CFR Part 141.41 (c) respectively, and may be used to meet the total simulated flight time allowed in 14 CFR Part 141, Appendices B through G and Appendix I.

1. Aerosim-Mechtronix Level 6 FTD
   The Aerosim-Mechtronix Level 6 flight training device (FTD) is an enclosed stationary trainer equipped with an exact replica of the aircraft that meets FAA Level 6 certification criteria. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.
2. Aerosim-Mechtronix Advanced Aviation Training Device (AATD)
   The Aerosim Model Aviator 100 Single Engine Land is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.
3. The Frasca International Inc. TruFlite Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.
4. The Redbird MCX Advanced Aviation Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically advanced, single and multi-engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.
5. The Frasca International Inc. RTD Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.

Jacksonville Facility (CRG)

Acron Aviation Academy's Jacksonville satellite base is located at the Jacksonville Executive at Craig Airport, 793 Amelia Earhart Drive, Jacksonville, FL 32225.

Chief and Designated Assistant Chief Flight Instructor

The Chief Flight Instructor and Assistant Chief Flight Instructor for Acron Aviation Academy is found in the current Webopss A006 and A007.

Airport

The Jacksonville Executive At Craig Airport (CRG) is the satellite base's main location for training. The airport has two (2) hard surface runways and meets the requirements of 14 CFR Part 141.38 for day and night flight operations.

An FAA contract control tower operates between 0600-2300 Monday through Friday and 0700- 2200 Saturday and Sunday.

The airport has ILS, VOR, and GPS approaches.

Fuel service is available from 0600-2200 hours daily. Maintenance service is available full time from 0700-1530 and on call 1530-2200 daily.

Training Airports

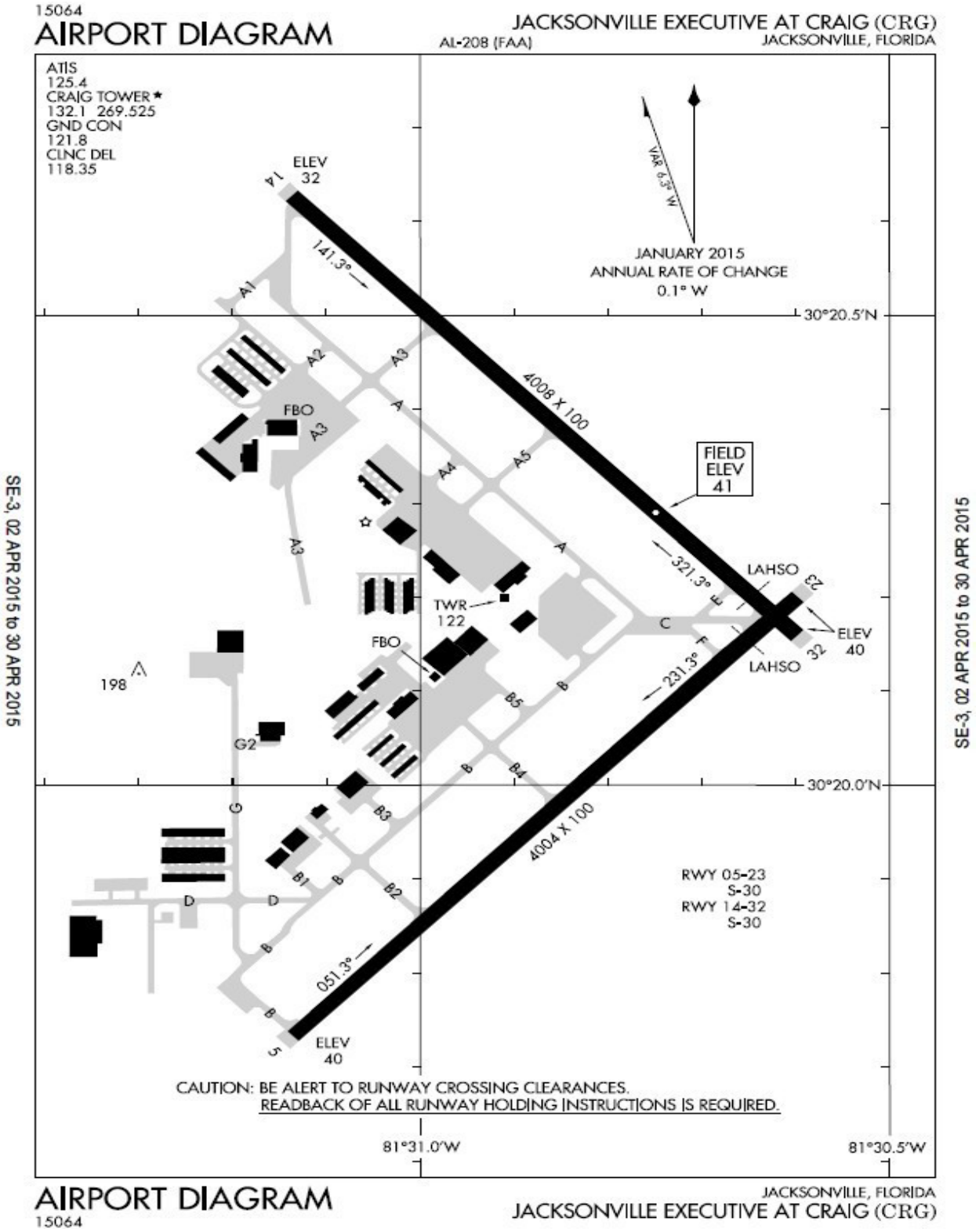
All training airports meet the requirements outlined in 14 CFR Part 141.38.

Direction is given to the Academy’s flight instructors through an Approved Airports list in supplemental documentation.

The Chief Flight Instructor or designated Assistant Chief Flight Instructor may approve any appropriate public use airport listed in the Airport/Facilities Directory.

Jacksonville/Crain Airport Diagram

*Note: Diagram not current and not to be used for navigation.



Operations Center

Acron Aviation Academy's Operations Center is the focal point for all flight lesson activity. Adequate areas have been provided for the dispatch and recovery of all flight lesson activity.

The Operations Center has adequate Pre-Post Flight Briefing Areas.

Pre-Post Flight Briefing Areas

Briefing Rooms 1 and 2 are equipped with tables and chairs adequate for both students and instructors. Briefing Room 1 may accommodate up to 16 students. Briefing Room 2 may accommodate up to 8 students. Briefing Room 2 may be reserved for FAA practical test oral examinations and stage checks as needed. All briefing areas are also equipped with erasable marker boards.

A telephone is available to provide communication to the nearest FAA Flight Service Station.

Simulator Hall

The Jacksonville facility has a Simulator Hall.The Sim Hall includes AATD simulators.

Equipment

Ground Training Aids

The training aids available for use, as appropriate, are as follows:

A. Audiovisual Aids
   Computers, Diagrams and Charts
B. Mockups
   Cockpit Procedures Trainer
C. Aircraft Components Flightinstruments

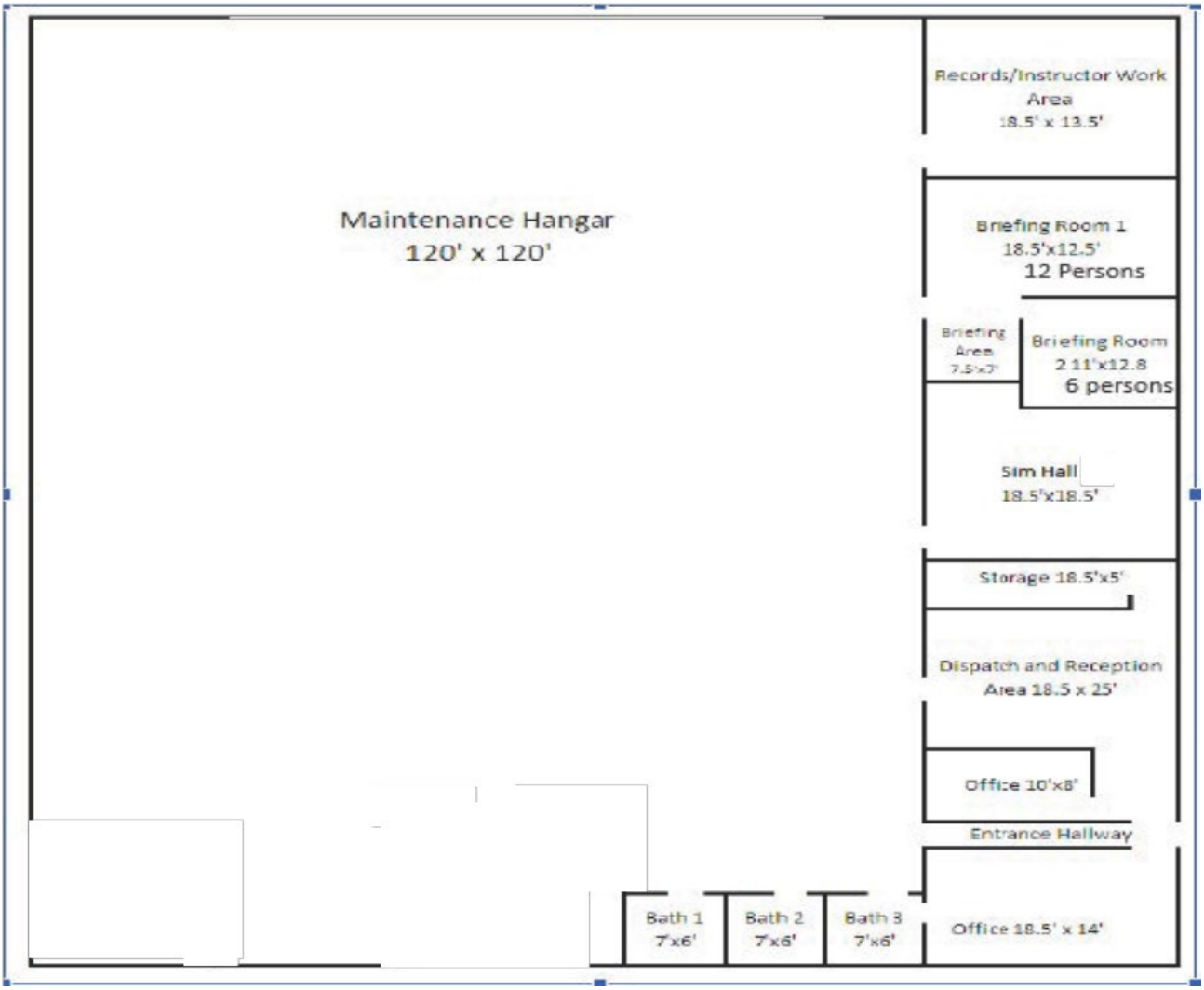
In addition, for enhancement and reinforcement of training a variety of video tapes and computer- based training is available.

Flight Training Devices

The Academy's flight training devices/approved aviation training devices meet the requirements specified in 14 CFR Part 141.41 (b)/14 CFR Part 141.41 (c) respectively, and may be used to meet the total simulated flight time allowed in 14 CFR Part 141, Appendices B through G and Appendix I.

-The Frasca International Inc. RTD Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane or multi engine(TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.

Operations Center Diagram



Academics

The Academics facilities are located on the campus of Jacksonville University. Located at 2800 University Blvd. N., Jacksonville, FL 32211.

Classrooms

Classroom D-153 in the Davis College of Business Building may accommodate 41 students. Depending on the size of the class and the scheduling needs of the college other rooms may be assigned.

Ground Training Aids

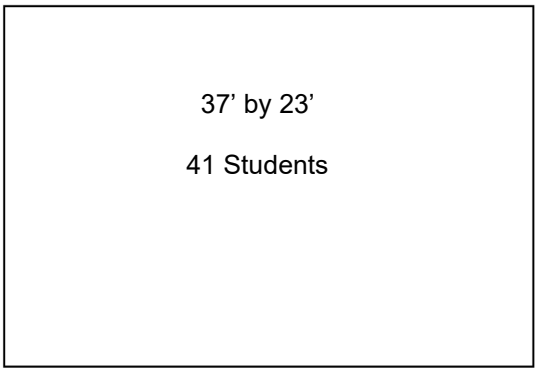
PC/VCR/DVD/Internet with overhead projection capability
Overhead projector
Computer-based training
Flight instruments
Static displays of aircraft engines and parts Aircraft models
Giant E6B White board

NOTE

All rooms and areas conform to local building, sanitation, and health codes. All rooms are designed and located so that students will not be distracted by the instruction conducted in other rooms or by other flight and maintenance operations on the airport.

Jacksonville University Classroom Diagram

Davis College of Business Room D-153



Chapter 2 – Commercial Airplane Single Engine Land (141.55(e)) Ground Syllabus

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Record of Revisions

REV NO.	DATE	INITIALS	LIST of CHANGES
Issue	04/28/2004	TS	
1	03/13/2012	RH	
2	11/04/2015	RH	
3	05/17/2017	NS	
4	05/17/2018	RH	
5	09/02/2020	NS	**Pages:1-28-**Name Change to L3Harris; Edited all reference to L3 Commercial Training Solutions to "L3Harris" Pg.2- Changed format to Record of revisions
6	04/07/2025	DK	Name change from L3Harris to Acron

After revision insertion, enter the date the revision is incorporated and initial in the appropriate column.

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List of Effective Pages

PAGE	REVISION	DATE	PAGE	REVISION	DATE
1	6	04/07/2025	2	6	04/07/2025
3	6	04/07/2025	4	6	04/07/2025
5	6	04/07/2025	6	6	04/07/2025
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21	6	04/07/2025	22	6	04/07/2025
23	6	04/07/2025	24	6	04/07/2025
25	6	04/07/2025	26	6	04/07/2025
27	6	04/07/2025	28	6	04/07/2025

Revision System

Commercial ASEL course revisions will be issued in consecutive numbers and each page of the new revision will contain the revision number and the date the revision was issued.

It is the responsibility of all affected personnel to periodically review the *Commercial ASEL course* to ensure that all current procedures are being followed.

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CASEL Course Objective

The objective of this course is to introduce the Pilot Under Instruction **(PUI)** to the Commercial ASEL aviation-training environment. The **PUI** will study all aeronautical knowledge areas required for a Commercial ASEL Pilot Certificate. **This course is designed for an applicant that already holds a Private Pilot Airplane Single Engine Land Certificate with Instrument Airplane Rating.**

The Instructor will conduct training using the following equipment:

Classroom
Computer
Projector
Whiteboard

Reference Textbooks for this course include (Not all inclusive):

14 CFR 1	AFM	POH/AFM
AC 60-28	FAA-H-8083-3	NOTAMS
AC 61-136	FAA-H-8083-15	
AC 91-74	FAA-H-8083-16	
AC 91.21-1	FAA-H-8083-25	
AC 120-108	FAA-ACS	

CASEL Course Completion Standards

At the completion of this course, the **PUI** will demonstrate to the Instructor a basic understanding of all aeronautical knowledge areas required for a Commercial ASEL Pilot Certificate. The **PUI** will be able to accomplish the objectives of the course to the appropriate level of performance, as outlined in the examination completion standards.

Suggested Course Time Allocation Table

	Lessons	Hours	Exam
1	Introduction to Commercial Piloting	1	
2	Human Factors	2	
3	Crew Resource Management	3	
4	Emergency Procedures	2	
5	Safety of Flight	1	
6	Commercial Maneuvers	1	
7	Systems	2	
8	Performance & Limitations	3	
9	Aeromedical Factors	3	
10	Airspace	3	
11	Meterology /Weather Services	3	
12	Cross-Country Flight Planning	3	
13	Stage I Review/Exam		2
	Stage I TOTALS:	28	2

STAGE II

Lesson			
14	Federal Aviation Regulations/NTSB	2.0	
15	Certificates and Documentation	2.0	
16	Advanced Aerodynamics	3.0	
17	High Altitude Operations	1.0	
18	Night Operations	1.0	
19	Weight and Balance	1.0	
20	STAGE II Review/Exam		2.0
21	Final Exam		3.0
Stage II Totals		10.0	5.0
CASEL Course Totals		**37.0**	**7 . 0**
	CAMEL Course Minimum	35.0	

Note: The individual times shown on this table are for instructor/student guidance only. They are not mandatory for each individual ground school lesson.

Stage 1

Stage Objective

The objective of this stage is to introduce the Pilot Under Instruction **(PUI)** to Aeromedical/Human Factors, CRM, Systems, Aerodynamics, Regulations, Weight and Balance, and Performance.

Stage Completion Standards

This stage is complete when the **PUI** has completed the Stage Exam with a minimum passing grade of 80 percent and the instructor has reviewed each incorrect response with the student to insure complete understanding.

Ground School Lesson 1: INTRODUCTION TO COMMERCIAL PILOTING

Lesson Objective

During this lesson, the **PUI** will have an introduction to Commercial Piloting.

Lesson Content

History of Airplanes Commercial Airplane Certificate Priv. Commercial Training	

Completion Standards

The lesson shall be complete when, by oral review, the PUI displays an understanding of the material covered.

Ground School Lesson 2: HUMAN FACTORS

Lesson Objective

During this lesson, the **PUI** will study human factors.

Lesson Content

History of Human Factors PIC Responsibilities Communication Resource Use Workload Management Situational Awareness	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 3: CREW RESOURCE MANAGEMENT

Lesson Objective

During this lesson, the **PUI** will study crew resource management.

Lesson Content

Checklist Usage Application Case Studies	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 4: EMERGENCY PROCEDURES

Lesson Objective

During this lesson, the **PUI** will study commercial flight considerations in the single engine environment.

Lesson Content

Emergency Approach and Landing Systems and Equipment Malfunctions Emergency Equipment and Survival Gear	
General Emergency Services Available to Pilots Distress and Urgency Procedures Two-Way Radio Communications Failure Aircraft Rescue and Fire Fighting Communications	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 5: SAFETY OF FLIGHT

Lesson Objective

During this lesson, the PUI will study safety of flight issues.

Lesson Content

Altimeter Setting Procedures Wake Turbulence Bird Hazards and Flight over National Refuges, Park, And Forests Potential Flight Hazards Safety, Accident, and Hazard Reports	

Completion Standards

The lesson shall be complete when, by oral examination, the **PUI** displays an understanding of the material covered.

Ground School Lesson 6: COMMERCIAL MANUEVERS

Lesson Objective

During this lesson, the **PUI** will study single engine commercial pilot maneuvers.

Lesson Content

Maximum Performance Takeoffs and Landings Steep turns Chandelles Lazy Eights Eights-on-Pylons Steep Spirals Power-off 180 Accuracy Approaches and Landings Emergency Descent	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 7: SYSTEMS

Lesson Objective

During this lesson, the **PUI** will study aircraft systems.

Lesson Content

Fuel Injection Systems Operating Procedures Engine Monitoring Turbocharging Systems Constant-Speed Propellers	
Ice Control Systems	
Landing Gear Systems Gear System Safety Operating Procedures	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 8: PERFORMANCE & LIMITATIONS

Lesson Objective

During this lesson, the **PUI** will study in depth the factors affecting airplane performance.

Lesson Content

Factors Affecting Performance Pilot Operating Handbook Performance Charts Effects of Exceeding Performance Limitations	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered

Ground School Lesson 9: AEROMEDICAL FACTORS

Lesson Objective

During this lesson, the **PUI** will study aeromedical factors.

Lesson Content

Hypoxia Hyperventilation Inner Ear and Sinus Spatial Disorientation Motion Sickness Carbon Monoxide Poisoning Stress and Fatigue Alcohol and Drugs Nitrogen	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 10: AIRSPACE

Lesson Objective

During this lesson, the **PUI** will study descriptions of and procedures for operating within the National Airspace System.

Lesson Content

VFR Minimums Airspace Classes A B C D E G Special Use Other	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered

Ground School Lesson 11: METEOROLOGY AND WEATHER SERVICES

Lesson Objective

During this lesson, the **PUI** will study meteorology and weather services. This review will include recognition of critical weather situations, wind-shear recognition and avoidance, and the use of aeronautical weather reports and forecasts.

Lesson Content

Wind Shear Thunderstorms Microbursts CAT Icing	
Reports, Charts, and Forecasts METAR, TAF Surface Analysis Radar Summary Winds and Temperatures Aloft Significant Weather Prognostic Convective Outlook AWOS, ASOS, ATIS Go/No Go Decision	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 12: CROSS-COUNTRY FLIGHT PLANNING

Lesson Objective

During this lesson, the **PUI** will study cross-country flight planning.

Lesson Content

Cross Country Flight Aeronautical Charts Airspace, Obstructions, and Terrain Checkpoints Altitudes Headings, Time, and Fuel Navigation and Communications Facilities NOTAMS and A/FD Navigation Log and Flight Plan	
High Altitude Operations Nighttime Operations	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 13: STAGE I REVIEW/EXAM

Lesson Objective

During this lesson, the **PUI** will be tested on comprehension of the academic material presented in Stage I.

Lesson Content

Stage I Review Stage I Exam

Completion Standards

The lesson shall be complete when, by written examination, the **PUI** displays an understanding of the material covered with a passing grade of at least 80%.

Stage 2

Ground School Lesson 14: FEDERAL AVIATION REGULATIONS/NTSB

Lesson Objective

During this lesson, the **PUI** will have an introduction to Commercial Piloting regulations/NTSB.

Lesson Content

Part 61 Part 91 Part 119 Part 121 Part 135 Part 141	
General Initial Notification of Aircraft Accidents, Incidents, and Overdue Aircraft Preservation of Aircraft Wreckage, Mail, Cargo, and Records Reporting of Aircraft Accidents, Incidents, and Overdue Aircraft	

Completion Standards

The lesson shall be complete when, by oral review, the PUI displays an understanding of the material covered.

Ground School Lesson 15: CERTIFICATES AND DOCUMENTS

Lesson Objective

During this lesson, the **PUI** will study certificates and documents.

Lesson Content

Commercial Pilot Certificate Medical Certificate Pilot Logbook Airworthiness and Registration POH/AFM Weight and Balance Data Equipment List	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 16: Basic AERODYNAMICS

Lesson Objective

During this lesson, the **PUI** will study airplane aerodynamics and the principles of flight.

Lesson Content

Lift Drag Thrust Weight and Load Factor The V-G Diagram Aircraft Stability Aerodynamics and Flight Maneuvers Stall and Spin Awareness	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 17: HIGH ATLITUDE OPERATIONS

Lesson Objective

During this lesson, the **PUI** will study high altitude operations.

Lesson Content

Oxygen Requirements Physiological Hazards Pressurization and Oxygen Systems Aviators Oxygen Care of Oxygen Bottles Rapid Decompression Cabin Pressurization Concepts Cabin Pressurization Systems	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 18: NIGHT OPERATIONS

Lesson Objective

During this lesson, the **PUI** will study night operations.

Lesson Content

Night Vision Night Illusions Interior Lights Flashlight Night Preflight Engine Starting/Aircraft Lights Taxi Orientation Takeoff and Climb In-flight Orientation Night BAI Night Emergencies Traffic Patterns Approaches and Landings Go-Around	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 19: WEIGHT & BALANCE

Lesson Objective

During this lesson, the **PUI** will study the factors affecting weight and balance.

Lesson Content

Weight Terms Effects of Too Much Weight Maneuvering Speed Center of Gravity Normal Loading Effects of Forward or Aft CG Weight Shifting Transport Loading	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 20: STAGE 11 REVIEW/EXAM

Lesson Objective

During this lesson, the **PUI** will be tested on comprehension of the academic material presented in Stage 11.

Lesson Content

Stage II Review Stage II Exam

Completion Standards

The **PUI** should complete the Commercial Stage II Exam with a minimum passing score of 80% and the instructor should review each incorrect response to ensure complete understanding.

Ground School Lesson 21: FINAL EXAM

Lesson Objective

During this lesson, the **PUI** will be tested on their comprehension of the academic material presented in this course.

Lesson Content

Final Exam

Completion Standards

The **PUI** should complete the Commercial End-of-Course Exam with a minimum passing score of 80% and the instructor should review each incorrect response to ensure complete understanding.

The student will demonstrate through written tests and school records that the aeronautical knowledge necessary to pass the FAA Commercial Pilot Airplane Knowledge Examinations has been acquired. The student must meet the requirements specified in 14 CFR Part 141, Appendix D.

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Chapter 3 - Commercial ASEL Part 141.55(E) Training Course Syllabus

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Record of Revisions

REV NO.	DATE	INITIALS	LIST of CHANGES
10	03/13/2012	SP	
11	08/03/2012	TM	
12	06/16/2016	JL	
13	05/17/2017	NS	
14	05/15/2018	CG	
15	08/30/2018	CG	
16	09/02/2020	NS	**Pages:1-52-**Name Change to L3Harris;Edited all reference to L3 Commercial Training Solutions to "L3Harris"
17	05/25/2021	CG	**Page: 1-66-** REV 16 was a name change; in error the name change update was made to a copy of REV 14, not 15.
18	04/07/2025	DK	Name change from L3harris to Acron

After revision insertion, enter the date the revision is incorporated and initial in the appropriate column.

Revision System

Commercial ASEL FAR Part 141.55(e) Syllabus revisions will be issued in consecutive numbers; and each page of the new revision will contain the revision number and the date the revision was issued.

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List of Effective Pages

	Commercial ASEL Syllabus				
PAGE	REVISION	DATE	PAGE	REVISION	DATE
1	18	04/07/2025	2	18	04/07/2025
3	18	04/07/2025	4	18	04/07/2025
5	18	04/07/2025	6	18	04/07/2025
7	18	04/07/2025	8	18	04/07/2025
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59	18	04/07/2025	60	18	04/07/2025
61	18	04/07/2025	62	18	04/07/2025
63	18	04/07/2025	64	18	04/07/2025
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Time Allocation Table

Stage	Lesson Number	Schedule Time	Ground Brief/ Oral	Pre Brief	Post Brief	FTD	Aircraft	Dual	Solo/ P IC	X-C	Night	Instrument Training	Stage Check	Lesson Equipment
1	1-GB	2.0	2.5											Ground
1	2-DFT	2.0		0.3	0.2	1.5		1.5				0.2		AATD/FTD
1	3-DL	2.0		0.3	0.8		2.0	2.0				0.2		AIRPLANE
1	4-GB	2.0	3.0											Ground
1	5-GB	2.0	3.0											Ground
1	6-DFT	2.0		0.3	0.5	2.0		2.0				1.8		AATD/FTD
1	7-DL	2.5		0.3	0.5		2.5	2.5				1.8		AIRPLANE
1	8-GB	3.0	3.0											Ground
1	9-DXC	3.5		0.3	0.3		3.5	3.5		3.5		2.5		AIRPLANE
1	10-DXCN	3.5		0.3	0.3		3.0	3.0		3.0	3.0	2.5		AIRPLANE
1	11-DL	2.5		0.3	0.3		3.5	3.5				0.5		AIRPLANE
1	12-STO	3.0	3.0										2.5	Ground
1	13-STF	3.0		0.3	0.8		2.5	2.5				0.5	2.5	AIRPLANE
2	14-DXC	8.0		0.3	0.3		7.5		7.5	7.5				AIRPLANE
2	15-DLN	3.0		0.3	0.3		2.5		2.5		2.5			AIRPLANE
2	16-DLN	3.0		0.3	0.3		2.5		2.5		2.5			AIRPLANE
2	17-GB	2.0	3.0											Ground
2	18-DL	2.5		0.3	0.3		2.5	2.5						AIRPLANE
2	19-DL	2.5		0.3	0.3		2.5	2.5						AIRPLANE
2	20-DL	2.5		0.3	0.3		2.5	2.5						AIRPLANE
2	21-DL	3.0		0.3	0.3		2.5	2.5						AIRPLANE
2	22-GB	2.0	3.0											Ground
2	23-DL	3.0		0.3	0.5		3.0	3.0						AIRPLANE
2	24-STO	3.0	3.0										2.5	Ground
2	25-STF	3.0		0.5	0.7		3.0	3.0					3.0	AIRPLANE
		TOTALS	23.5	4.5	6.5	3.5	46.0	37.0	12.5	14.5	8.0	10.0	10.5	

Course Minimums

MINIMUM BRIEFING	DUAL COMPLEX/TAA	SPIC	SPIC NIGHT	SPIC TOWERED NIGHT TAKEOFFS AND LANDINGS	TOTAL INSTUMENT
8.25	10 (8 in	1	5	10	10 (8 in airplane)

Cross Country (100 nm night)	2 hours
Cross Country (100 nm day)	2 hours

Stage 1

Stage Objective

The objective of this stage is to introduce the Pilot Under Instruction (**PUI**) to VFR/IFR Cross Country operations in preparation for the required solo/ supervised pilot in command flight lessons. The **PUI** will be introduced to cross-country missions and will learn the skills necessary to perform cross-country flights using pilotage and dead reckoning, en-route diversion procedures, and lost procedures, while incorporating the use of navigation systems and radar services. Additionally, the **PUI** will be introduced to single engine flight maneuvers and instrument skills and procedures. The **PUI** will continue using aeronautical decision-making skills that will set the foundation for all future aviation training.

The **PUI** will conduct training in the following equipment:

Advanced Aviation Training Device (**AATD**)
Flight Training Device (**AATD/FTD**)

Single Engine Airplane

Stage Completion Standards

At the completion of this stage, the **PUI** will demonstrate to the Instructor Pilot (**IP**) an understanding of cross-country operations, single engine flight maneuvers, and instrument skills and procedures. The **PUI** will be able to accomplish the objectives of Stage 1 to the appropriate level of performance, as outlined in the lesson completion standards.

NOTE

*In the event an **AATD/FTD** is not available, the training event prescribed can be conducted in an **aircraft.***

NOTE

*All flight events in this course can be conducted in any single engine aircraft in accordance with Chapter 1 of this Training Course Outline. However, the student **must have a minimum 10 hours of training in** a **complex airplane** or **technically advanced airplane** that meets the requirements of 61.129 (j) or any combination thereof.*

Lesson 1- GB

Equipment

Briefing Room

Lesson Objective

During this lesson, the **PUI** will be introduced to Pilot Qualifications, operational policies and procedures, the Training Course Outline, normal procedures of a single engine airplane, and Crew Resource Management (CRM).

Lesson Content

Training Record	
Pilot Qualification	Introduction
Identity Verification/ TSA Clearance (As Required)	Review
Copy Photo Identification	Review
Copy Pilot Certificate	Review
Copy Medical Certificate	Review
Create Training Folder	Review
Stage Goals and Objective	
Stage 1 Overview	Introduction
Publications	
Acron Aviation Operations Manual	Review
Acron Aviation Standardizations Manual	Review
Training Course Outline	Review
Pilot Operating Handbook	Introduction
Airman Certification Standards (ACS)-Commercial Pilot	Introduction
Checklist Usage	
Normal Procedures Checklist	Introduction
Chair flying	
Use/Location of Equipment	Introduction
Cockpit Flows	Introduction
CRM Practice	Introduction
Crew Resource Management	
Roles and Responsibilities	Introduction
Performance and Limitations	
Factors Affecting Performance	Review
Aerodynamics	Review
Take Off Data Card/ Weight and Balance	Review
Additional Areas	
As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this Lesson, the **PUI** will have increased knowledge of normal procedures of a single engine airplane. During the post flight debrief the **PUI** will be assigned specific tasks to prepare for the next lesson.

Lesson 2-DFT

Equipment

AATD/FTD

Lesson Objective

This Lesson provides an introduction to steep turns and emergency/abnormal operations and a review of the procedures and skills needed in a single engine airplane. All tasks will be conducted using Crew Resource Management (CRM) techniques and procedures.

Lesson Content

CRM	
Risk Assessment	Introduction
Collision Avoidance	Introduction
Task Management	Introduction
Automation Management (If Installed)	Introduction
Preflight Procedures	
Lesson Planning	Introduction
Preflight Briefing	Introduction
Cockpit Familiarization	
Flight Controls	Introduction
Engine Controls	Introduction
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Introduction
Emergency Procedures Checklist	Introduction
Flows	
Cockpit Flows	Introduction
Engine Start	
General Starting Procedures	Introduction
Engine Fire On Start Procedures	Introduction
Operation of Systems In Flight	
Powerplant Operations	Introduction
Use of Constant Speed Propeller (As Applicable)	Introduction
Use of Retractable Landing Gear (As Applicable)	Introduction
Use of Cowl Flaps (As Applicable)	Introduction
Mixture Leaning including use of EGT	Introduction

Before Takeoff Check	
Engine Check	Introduction
Instrument Check	Introduction
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff Procedures	Introduction
Aborted Takeoff	Introduction
Operation of Systems In-Flight Powerplant Operation	Introduction
Slow Flight	
Pitch/Airspeed Control	Introduction
Power/Altitude Control	Introduction
Climbs/Turns/Descents	Introduction
Stalls	
Power-Off Stalls	Introduction
Power-On Stalls	Introduction
Accelerated Stalls	Introduction
Steep Turns	
Steep Turns	Introduction
Unusual Attitudes (IR)	
Unusual Attitudes/Recovery Procedures	Introduction
Emergency/ Abnormal Procedures	
Engine Failure in Flight	Introduction
Systems and Equipment Malfunctions	Introduction
Emergency Descent	Introduction
Approach and Landing Procedures	
Normal and/or Crosswind Approach Procedures	Introduction
Go-Around/Rejected Landing	Introduction
Postflight	
Operations Postflight Debrief	Introduction
Special Emphasis Areas	
Collision Avoidance	Introduction
Positive Exchange of Flight Controls	Introduction
Aeronautical Decision Making (ADM)	Introduction
Stall/Spin Awareness	Introduction

Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this Lesson the **PUI** will have gained an increased proficiency of basic aircraft avionics, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 20 degrees, altitude will be maintained within +/- 200 feet, and airspeed will be maintained within +/-10 kts during maneuvers.

Landings will be made in accordance with Commercial Airman Certification Standards with the following modifications: Airspeed -5/+10 knots, and touchdown within -0/+400 feet of a specified point.

During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 3-DL

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

This lesson provides a review of the procedures and skills needed in a single engine airplane. During this lesson, the **PUI** will be introduced to emergency operations in an airplane and commercial maneuvers and procedures.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Introduction
Emergency Procedures Checklist	Introduction
Flows	
Cockpit Flows	Review/Introduction
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Introduction
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Aborted Takeoff	Review

Operation of Systems In-Flight Powerplant Operation	Review
Slow Flight Pitch/Airspeed Control Power/Altitude Control Climbs/Turns/Descents	 Review Review Review
Stalls Power-Off Stalls Power-On Stalls Accelerated Stalls	 Review Review Review
Operation of Systems In Flight Powerplant Operations Use of Constant Speed Propeller (As Applicable) Use of Retractable Landing Gear (As Applicable) Use of Cowl Flaps (As Applicable) Mixture Leaning including use of EGT	 Review Review Review Review Review
Emergency/Abnormal Procedures Engine Failure in Flight Systems and Equipment Malfunctions Emergency Descent	 Review Review Review
Steep Turns Steep Turns	Review
Unusual Attitudes (IR) Unusual Attitudes/Recovery Procedures	Review
Approach and Landing Procedures Normal and/or Crosswind Approach Procedures Go-Around/Rejected Landing	 Review Review
Postflight Operations Postflight Debrief	Review
Special Emphasis Areas Stall/Spin Awareness Land and Hold Short Operations (LAHSO) Aeronautical Decision Making (ADM)	 Review Review Review

Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained an increased proficiency of basic aircraft avionics, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 15 degrees, altitude will be maintained within +/- 150 feet, and airspeed will be maintained within +/-10 kts during maneuvers.

Landings will be made in accordance with the Commercial Airman Certification Standards with the following modifications: Airspeed -5/+10 knots, and touchdown within -0/+300 feet of a specified point.

During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 4-GB

Equipment

Briefing Room

Lesson Objective

During this Lesson, the **PUI** will be introduced to the regulations pertaining to Commercial Pilots, high altitude operations and pressurization.

Lesson Content

Regulations	
Part 1	Introduction
Part 39	Introduction
Part 43	Introduction
Part 61	Introduction
Part 71	Introduction
Part 73	Introduction
Part 91	Introduction
Part 119	Introduction
Advisory Circulars	Introduction
Commercial Pilot Privileges & Limitations	Introduction
Common Carriage	Introduction
Private Carriage	Introduction
Holding Out	Introduction
Pressurization and Supplemental Oxygen	
Physiological Factors	Introduction
Impairment	Introduction
Hypoxia	Introduction
Time of Useful Consciousness (TUC)	Introduction
Continuous-Flow Oxygen System	Introduction
Diluter-Demand Oxygen System	Introduction
Pressure-Demand Oxygen System	Introduction
“Aviator’s Breathing Oxygen”	Introduction
Risk Management	Introduction
Regulatory Requirements (Pressurized and Non-Pressurized)	Introduction
Checklist Usage	
Emergency Procedures Checklist	Review/Introduction
Special Emphasis Areas	
Controlled Flight into Terrain	Review
Runway Incursion Avoidance	Review

Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have an increased level of knowledge in all areas listed in the table above, and all necessary items on the emergency checklist memorized. During the post flight debrief the **PUI** will be assigned specific tasks to prepare for the next lesson.

Lesson 5-GB

Equipment

Briefing Room

Lesson Objective

During this Lesson, the **PUI** will review basic attitude instrument flying procedures.

Lesson Content

Holding Procedures	
Holding Procedures	Review
Entries	Review
Speeds/Altitudes	Review
Timing/Patterns	Review
Instrument Approach Procedures	
Approach Briefing	Introduction
Precision Approaches	Introduction
Non-Precision Approaches	Introduction
Circling Approach Procedures	Introduction
Weather Theory	
Atmosphere- Composition and Stability	Review
Temperature	Review
Atmospheric Pressure	Review
Wind	Review
Moisture, Clouds, and Precipitation	Review
Stable and Unstable Air	Review
Weather System Formation-Air Masses and Fronts	Review
Turbulence	Review
Icing and freezing level information	Review
Fog	Review
Frost	Review
Thunderstorms and Microbursts	Review
Special Emphasis Areas	
Positive Aircraft Control	Review
Controlled Flight into Terrain	Review
Additional Areas	
As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this Lesson, the **PUI** will have knowledge of the tasks that meets or exceeds the outcome level specified with respect to the Instrument Pilot Standards.

Lesson 6-DFT

Equipment

AATD/FTD

Lesson Objective

This Lesson provides a review of basic attitude instrument flying in a single engine airplane.

Lesson Content

CRM Risk Assessment Collision Avoidance Task Management Automation Management (If Installed)	 Review Review Review Review
Preflight Procedures Lesson Planning Preflight Briefing	 Review Review
Cockpit Familiarization Flight Controls Engine Controls	 Review Review
Checklist Usage Normal Procedures Checklist	 Review
Flows Cockpit Flows	 Review
Engine Start General Starting Procedures	 Review
Before Takeoff Check Instrument Check	 Review
Takeoff and Climb Procedures Instrument Takeoff	 Review
Operation of Systems In-Flight Powerplant Operation	 Review
Flight by Reference to Instruments	
Climbs (IR) Constant Airspeed Climbs Constant Rate Climbs	 Introduction Introduction

Operation of Systems In Flight	
Powerplant Operations	Review
Use of Constant Speed Propeller (As Applicable)	Review
Use of Retractable Landing Gear (As Applicable)	Review
Use of Cowl Flaps (As Applicable)	Review
Turns (IR)	
Standard Rate Turns	Introduction
Compass Turns	Introduction
Timed Turns to Magnetic Headings	Introduction
Steep Turns (IR)	
Steep Turns	Introduction
Unusual Attitudes (IR)	
Unusual Attitudes/Recovery Procedures	Introduction
Descents (IR)	
Constant Airspeed Descents	Introduction
Constant Rate Descents	Introduction
Holding Procedures	
Holding Procedures	Introduction
Hold	Introduction
Instrument Approach Procedures	
ILS Approach & Procedures	Introduction
Partial Panel LOC Approach & Procedures	Introduction
VOR Approach & Procedures	Introduction
Approach and Landing Procedures	
Normal and/or Crosswind Approach Procedures	Review
Postflight Operations	
Postflight Debrief	Review
Special Emphasis Areas	
Controlled Flight into Terrain (CFIT)	Review
Positive Aircraft Control	Review
Stall/Spin Awareness	Review
Checklist Usage	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson, the student will have met the desired outcome level for each task with respect to the FAA Instrument Rating Standards.

During the post flight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 7-DL

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

This Lesson provides a review of intercepting procedures, tracking procedures, and a review of full and partial panel instrument approach procedures in an airplane.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Short-Field Takeoff	Introduction
Operation of Systems In Flight	
Powerplant Operations	Review
Use of Constant Speed Propeller (As Applicable)	Review
Use of Retractable Landing Gear (As Applicable)	Review
Use of Cowl Flaps (As Applicable)	Review

Turns (IR) Standard Rate Turns	Review
Steep Turns (IR) Steep Turns	Review
Unusual Attitudes (IR) Unusual Attitudes/Recovery Procedures	Review
Holding Procedures Holding Procedures Hold	 Review Review
Instrument Approach Procedures ILS Approach & Procedures Partial Panel Non-Precision Approach & Procedures Non-Precision Approach & Procedures	 Introduction Introduction Introduction
Approach and Landing Procedures Normal and/or Crosswind Approach Procedures Short Field Landing	 Review Introduction
Postflight Operations Postflight Debrief	Review
Special Emphasis Areas Positive Aircraft Control Controlled Flight Into Terrain (CFIT) Checklist Usage	 Review Review Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson, the **PUI** will demonstrate knowledge of the tasks listed, as outlined in the preceding table with respect the Instrument Pilot Standards.

Landings will be made in accordance with Commercial Airman Certification Standards with the following modifications: Airspeed -5/+10 knots, for normal landings touchdown within -0/+300 feet of a specified point, and for short field landings touchdown within - 0/+200 feet of a specified point.

During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 8-GB

Equipment

Briefing Room

Lesson Objective

This Lesson provides a review of cross country flight planning, weather information, night flight, and aeromedical factors. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance required to complete the Lesson.

Lesson Content

Weather Information	
Graphical Forecasts for Aviation (GFA)	Review
Terminal Aerodrome Forecast (TAF)	Review
Aviation Routine Weather Report (METAR)	Review
Wind and Temperature Aloft Forecast (FB)	Review
Airmets (WA), Sigmets (WS), and Convective Sigmets (WST)	Review
IFR and VFR Cross-Country Planning	
Plan Cross-Country	Review
Cross-Country Navigation Log	Review
Route Planning	
Most Economical Route Possible	Review
Most Efficient Route Possible	Review
Elements of a Flight Plan	Review
Time, Speed, Distance, Heading, ETA	Review
Airspace Consideration	Review
Altitude Selection	Review
Procedures to Open and Close a VFR Flight Plan	Review
PAVE Checklist	Review
Fuel Requirements	Review
Improper Fuel Planning	Review
ATC Limitations	Review
National Airspace System	
Controlled Airspace	Review
Uncontrolled Airspace	Review
Special Use Airspace	Review
Other Airspace	Review

Night Flight	
Planning for Night flights	Review
Oxygen Requirements/Recommendations	Review
Night Vision/Scanning	Review
Cockpit Lighting	Review
Human Factors	
Aeromedical and Physiological Issues	
In-Flight Illusions	Review
Landing Illusions	Review
Night Illusions	Review
Spatial Disorientation	Review
Special Emphasis Areas	
All Areas	Review
Additional Areas	
As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson, the **PUI** will have the knowledge to properly plan a VFR and IFR cross-country flight. Additionally the **PUI** must have knowledge of the tasks listed in the table above that meets or exceeds the standards outlined in the current Commercial Pilot Airman Certification Standards. During the post flight debrief the **PUI** will be assigned specific tasks to prepare for the next lesson.

Lesson 9-DXC

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

This lesson provides a review of the procedures and skills needed in a single engine airplane and fulfills the requirements of Part 141, Appendix D 4(b)(1)iii and Part 61.129(a)(3)iii. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance required to complete the lesson.

The **PUI** will plan and execute a DAY IFR cross-country to an airport more than 100 NM straight line distance from the original point of departure and at least 2.0 hours cross country time in duration.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Personal Weather Minimums Checklist	Review
Automation Management (if installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Preflight Inspection	Review
Route	
Most Economical Route Possible	Review
Most Efficient Route Possible	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review

Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel or Castering Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Climbs	
Constant Rate Climbs	Review
Constant Airspeed Climbs	Review
Operation of Systems In Flight	
Powerplant Operations	Review
Use of Constant Speed Propeller (As Applicable)	Review
Use of Retractable Landing Gear (As Applicable)	Review
Use of Cowl Flaps (As Applicable)	Review
Cross Country Flight (IR)	
Landing at an Airport more than 100 nm distance	Execute
Cross Country 2 hours in duration	Execute
Contact FSS for Weather Update	Introduction
Descents	
Constant Rate Descents	Review
Constant Airspeed Descents	Review
Instrument Approach Procedures	
ILS Approach & Procedures	Review
Non-Precision Approach & Procedures	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Postflight Operations	
Postflight Debrief	Review
Postflight Inspection	Review
Special Emphasis Areas	
Collision Avoidance	Review
Runway Incursion Avoidance	Review
Controlled Flight Into Terrain (CFIT)	Review

Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson, the **PUI** will have completed a DAY cross-country flight to an airport more than 100 nm straight-line distance from the original point of departure and 2 hours in cross-country duration. The **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures; aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table with respect to the Commercial Pilot ACS.

During the instrument approaches, the **PUI** will have met the desired outcome level for each task with respect to the FAA Instrument Rating Standards. Commercial Pilot ACS will be maintained for each takeoff and landing.

During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 10-DXCN

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

This night lesson provides a review of the procedures and skills needed in a single engine airplane and fulfills the requirements of Part 141, Appendix D 4(b)(1) iv and Part 61.129(a)(3) iv,. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance to complete the lesson.

The **PUI** will plan and execute a NIGHT IFR cross-country to an airport more than 100 NM straight line distance from the airport of departure and at least two hours cross country time in duration.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Personal Weather Minimums Checklist	Review
Automation Management (if installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Inspection	Review
Preflight Briefing	Review
Route	
Most Economical Route Possible	Review
Most Efficient Route Possible	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review

Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel or Castering Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Emergency/Abnormal Procedures Systems and Equipment malfunctions	Review
Operation of Systems In Flight	
Powerplant Operations	Review
Use of Constant Speed Propeller (As Applicable)	Review
Use of Retractable Landing Gear (As Applicable)	Review
Use of Cowl Flaps (As Applicable)	Review
Cross Country Flight(IR)	
Landing at an Airport more than 100 nm distance	Review
Cross Country 2 hours in duration	Review
Contact FSS for Weather Update	Review
Instrument Approach Procedures	
ILS Approach & Procedures	Review
Non-Precision Approach & Procedures	Review
Approach and Landing Procedures	
Short-field Landing	Review
Normal and/or Crosswind Landing	Review
Postflight Operations	
Securing Aircraft	Review
Postflight Inspection	Review
Postflight Debrief	Review
Special Emphasis Areas	
Collision Avoidance	Review
Runway Incursion Avoidance	Review
Controlled Flight Into Terrain (CFIT)	Review

Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson, the **PUI** will have completed a NIGHT cross-country flight to an airport more than 100 nm straight-line distance from the original point of departure and 2 hours in cross-country duration. The **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures; aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table with respect to the Commercial Pilot ACS.

During the instrument approaches, the **PUI** will have met the desired outcome level for each task with respect to the FAA Instrument Rating Standards. Commercial Pilot ACS will be maintained for each take-off and landing.

During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 11-DL

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

This Lesson provides a review of the procedures and skills needed in a single engine airplane.

For this lesson, the **PUI** will prepare a cross-country navigation log to demonstrate proper VFR navigation procedures to include E6B Usage, Pilotage, dead reckoning and diversion procedures.

All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Personal Weather Minimums Checklist	Review
Automation Management (if installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Inspection	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Review

Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel or Castering Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short Field Takeoff	Review
VFR Navigation	
Pilotage	
Dead Reckoning	Review
E6B Usage	Review
Diversion Procedures	Review
Open VFR Flight Plan (Simulate)	Review
Emergency/Abnormal Procedures	
Engine Failure In Flight	Review
Systems and Equipment Malfunctions	Review
Steep Turns	
Steep Turns	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Stalls	
Power-Off Stalls	Review
Power-On Stalls	Review
Accelerated Stalls	Review
Instrument Approach Procedures	
ILS Approach & Procedures	
Non-Precision Approach & Procedures	Review
Unusual Attitudes(IR)	
Recovery from Unusual Attitudes	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short Field Landing	Review
Go-Around/Rejected Landing	Review

Postflight Operations Securing Aircraft Postflight Inspection Postflight Debrief	 Review Review Review
Special Emphasis Areas All Areas	 Review
Additional Areas As Assigned By Instructor (If Necessary)	 Review

Completion Standards

At the completion of this lesson, the **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed in the table above. Heading must be maintained within +/- 10 degrees, airspeed must be maintained within+/- 10 KIAS, and altitude must be maintained within +/- 100 feet.

During the instrument approaches student will have met the desired outcome level for each task with respect to the FAA Instrument Rating Standards and Commercial Pilot ACS will be maintained for each maneuver, take-off and landing. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 12-STO-Stage Check (Oral)

Equipment

Briefing Area

Lesson Objective

This lesson will be conducted by the Chief/Assistant Chief or Designated Check Pilot.

The **IP** will evaluate the **PUI** during the oral. This lesson provides an evaluation of knowledge related to pilot qualifications/certificates and documents, airworthiness requirements, weather information, VFR cross country flight planning, national airspace system, performance and limitations, operations of systems, human factors/aeromedical factors, principles of pressurization, and supplemental oxygen.

Oral Pilot Qualifications/Certificates and Documents Airworthiness Requirements Weather Information VFR Cross Country Flight Planning National Airspace System Performance and Limitations Human Factors/Aeromedical Factors Principles of Pressurization Supplemental Oxygen
Special Emphasis Areas All Areas
Additional Areas As Assigned By Instructor (If Necessary)

Completion Standards

At the completion of this lesson, the **PUI** will have knowledge of the tasks that meets or exceeds the outcome level specified with respect to the Commercial Pilot ACS.

Lesson 13-STF-Stage Check (Flight)

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

This lesson provides a review of the flight maneuvers and procedures introduced during stage 1. This lesson will be conducted by the Chief/Assistant Chief or Designated Check Pilot. All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance to complete the lesson.

Lesson Content

SRM Risk Assessment Collision Avoidance Task Management Personal Weather Minimums Checklist Automation Management (If Installed)
Preflight Procedures Lesson Planning Preflight Inspection Preflight Briefing
Route Most Economical Route Possible Most Efficient Route Possible
Cockpit Familiarization Flight Controls Engine Controls
Checklist Usage Normal Procedures Checklist Abnormal Procedures Checklist Emergency Procedures Checklist
Flows Cockpit Flows
Engine Start General Starting Procedures Engine Fire On Start Procedures

Taxi Operations Basic Aircraft Control Nose Wheel or Castering Nose Wheel Steering Differential Braking
Before Takeoff Check Engine Check Instrument Check
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff Short-Field Takeoff
Operation of Systems In Flight Powerplant Operations Use of Constant Speed Propeller (As Applicable) Use of Retractable Landing Gear (As Applicable) Use of Cowl Flaps (As Applicable)
Emergency/Abnormal Procedures Engine Failure In Flight Systems and Equipment Malfunctions
Steep Turns Steep Turns
Slow Flight Pitch/Airspeed Control Power/Altitude Control
Stalls Power-Off Stalls Power-On Stalls Accelerated Stalls
Unusual Attitudes(IR) Recovery from Unusual Attitudes
VFRNavigation Pilotage Dead Reckoning E6B Usage DiversionProcedures Open VFR Flight Plan (Simulate)

Instrument Approach Procedures ILS Approach & Procedures Non-Precision Approach & Procedures
Approach and Landing Procedures Normal and/or Crosswind Landing Short-field Landing Go-Around/Rejected Landing
Postflight Operations Securing Aircraft Postflight Inspection Postflight Debrief
Special Emphasis Areas All Areas
Additional Areas As Assigned By Instructor (If Necessary)

Completion Standards

At the completion of this lesson, the **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading must be maintained within +/- 10 degrees, airspeed must be maintained within +/- 10 KIAS, and altitude must be maintained within +/- 100 feet.

During the instrument approaches student will have met the desired outcome level for each task with respect to the FAA Instrument Rating Standards and Commercial Pilot ACS will be maintained for each maneuver, take-off and landing.

Additionally the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Stage 2

Stage Objective

The objective of this stage is to introduce the pilot under instruction (**PUI**) to commercial flight maneuvers and to further develop the **PUI's** VFR navigational skills traffic pattern operations, and Single Pilot Resource Management (SRM) skills in all areas of flight.

The **PUI** will conduct training in the following equipment:

Advanced Aviation Training Device (**AATD**)
Flight Training Device (**AATD/FTD**)

Single Engine Airplane

Stage Completion Standards

At the completion of this stage, the **PUI** will demonstrate to the **IP** an increased level of proficiency in VFR navigation procedures, commercial flight maneuvers, and traffic pattern operations. Additionally the **PUI** will be exhibiting SRM skills in all phases of flight with little to no input from the **IP**. The **PUI** will be able to accomplish the objectives of Stage 2 to the appropriate level of performance, as outlined in the Lesson completion standards.

NOTE

*In the event an **AATD/FTD** is not available the training event prescribed can be conducted in an **aircraft***

NOTE

*All flight events in this course can be conducted in any single engine aircraft in accordance with Chapter 1 of this Training Course Outline. However, the student **must have a minimum 10 hours of training** in a **complex airplane** or **technically advanced airplane** that meets the requirements of 61.129 (j) or any combination thereof.*

Lesson14-DXC/SXC

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

During this lesson, the **PUI** will conduct a solo flight or will be performing the duties of pilot in command while under the supervision of a certificated flight instructor. This fulfills the requirements of Part 141 Appendix D 5(a) (2) and, impart, the requirements of Part 141 Appendix D 5 (a). All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance to complete the lesson.

The **PUI** will plan a multi segment cross-country flight with landings at a minimum of three (3) points, and one segment of the flight consisting of a straight-line distance of at least 250 NM.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Personal Weather Minimums Checklist	Review
Task Management	Review
Automation Management (if installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Inspection	Review
Preflight Briefing	Review
Route	
Most Economical Route Possible	Review
Most Efficient Route Possible	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review

Flows Cockpit Flows	 Review
Engine Start General Starting Procedures Engine Fire On Start Procedures	 Review Review
Taxi Operations Basic Aircraft Control Nose Wheel or Castering Nose Wheel Steering Differential Braking	 Review Review Review
Before Takeoff Check Engine Check Instrument Check	 Review Review
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff Short-Field Takeoff Soft-Field Takeoff	 Review Review Review
Traffic Patterns Traffic Pattern Operations ATC Light Gun Signals	 Review Review
Emergency/Abnormal Procedures Systems and Equipment Malfunctions	 Review
Cross Country Flight One XC leg more than 250 nm from airport of departure Landings at three points Open VFR Flight Plan	 Execute Execute Review
Approach and Landing Procedures Normal and/or Crosswind Landing Short-field Landing Soft-Field Landing Go-Around/Rejected Landing	 Review Review Review Review
Postflight Operations Securing Aircraft Postflight Inspection Postflight Debrief	 Review Review Review
Special Emphasis Areas All Areas	 Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have met the approved FAA Practical Test Completion Standards for the Commercial Pilot Certificate with Airplane single engine Class Rating privileges.

Lesson15-DLN/SLN

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

This night lesson the **PUI** will conduct a solo flight or will be performing the duties of pilot in command while under the supervision of a certificated flight instructor. This lesson provides a review of the procedures and skills needed for commercial pilot certification. This fulfills, in part, the requirements of Part 141 Appendix D 5(a) and Part 141 Appendix D 5(a) (3). All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures. Emphasis should be placed on destinations with active control towers.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Personal Weather Minimums Checklist	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures Lesson	
Planning	Review
Preflight Inspection	Review
Preflight Briefing	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel or Castering Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check Engine Check	
Instrument Check	Review
	Review
Takeoff and Climb Procedures Normal	
and/or Crosswind Takeoff	Review

Traffic Patterns Traffic Pattern Operations	Review
Emergency/Abnormal Procedures Engine Failure In Flight Systems and Equipment Malfunctions	Review Review
Steep Turns Steep Turns	Review
Approach and Landing Procedures Normal and/or Crosswind Landing Short-Field Landing Go-Around/Rejected Landing	Review Review Review
Postflight Operations Postflight Debrief Postflight Inspection	Review Review
Special Emphasis Areas Positive Aircraft Control Stall/Spin Awareness Checklist Usage	Review Review Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table with respect to the Commercial Pilot ACS. Additionally the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson16-DLN/SLN

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

This night lesson the **PUI** will conduct a solo flight or will be performing the duties of pilot in command while under the supervision of a certificated flight instructor. This lesson provides a
review of night operations. This fulfills in part the requirements of Part 141 Appendix D 5(a) and Part 141 Appendix D 5(a) (3). All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance to complete the Lesson.

At the completion of this lesson the **PUI** should have logged more than 5 hours night in VFR conditions in a single engine airplane solo or while performing the duties of pilot in command under the supervision of a flight instructor to include 10 takeoffs and 10 landings (each landing involving a flight with a traffic pattern), at an airport with an operating control tower.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Personal Weather Minimums Checklist	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Inspection	Review
Preflight Briefing	Review
Route	
Most Economical Route Possible	Review
Most Efficient Route Possible	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review

Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel or Castering Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Emergency/Abnormal Procedures	
Engine Failure In Flight	Review
Systems and Equipment Malfunctions	Review
Steep Turns	
Steep Turns	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-field Landing	Review
Soft-Field Landing	Review
Go-Around/Rejected Landing	Review
Postflight Operations Securing	
Aircraft	Review
Postflight Inspection	Review
Postflight Debrief	Review
Special Emphasis Areas	
All Areas	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** should have 5 hours of night in VFR conditions in a single engine airplane solo or while performing the duties of pilot in command under the supervision of a flight instructor to include 10 takeoffs and 10 landings (each landing involving a flight with a traffic pattern), at an airport with an operating control tower. The **PUI** will demonstrate knowledge of the tasks listed as outlined in the preceding table. All maneuvers and procedures will be conducted in accordance with the current FAA Commercial Pilot ACS. Additionally the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 17-GB

Equipment

Briefing Room

Lesson Objective

During this lesson, the **PUI** will be a review of weight and balance, performance, limitations, and preflight procedures pertaining to a single engine airplane.

Lesson Content

Weight & Balance	
Aircraft Weights	Review
Center of Gravity	Review
Arm	Review
Moment	Review
Center of Gravity Envelope	Review
Weight Shift Calculations/Formula	Introduction
Takeoff Data Card	
Pressure Altitude	Review
Density Altitude	Review
Fuel Requirements	Review
Takeoff & Landing Distances	Review
Aircraft Performance Data	
Performance Charts	Introduction
V Speeds	Introduction
Operations of Systems	
Powerplant	Introduction
Propeller	Introduction
Primary Flight Controls and Trim	Introduction
Fuel, Oil, and Hydraulic	Introduction
Checklist Test	
Checklist Test	Introduction
Commercial Flight Maneuvers	
Chandelles	
Lazy Eights	Introduction
Steep Spirals	Introduction
Eights On Pylons	Introduction
Special Emphasis Areas	Review
Checklist Usage	
Wake Turbulence Avoidance	Review

Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the Completion of this lesson, the **PUI** will have increased knowledge in all areas listed in the table above. During the post flight debrief the **PUI** will be assigned specific tasks to prepare for the next lesson.

Lesson 18-DL

Equipment

Single Engine Airplane

Lesson Objective

This lesson introduces commercial flight maneuvers and skills needed in a single engine airplane.

Lesson Content

SRM Risk Assessment Collision Avoidance Task Management Automation Management (As Applicable)	 Review Review Review Review
Cockpit Familiarization Flight Controls Engine Controls	 Review Review
Checklist Usage Normal Procedures Checklist	 Review
Flows Cockpit Flows	 Review
Engine Start General Starting Procedures	 Review
Taxi Operations Basic Aircraft Control Nose Wheel or Castering Nose Wheel Steering Differential Braking	 Review Review Review
Before Takeoff Check Engine Check Instrument Check	 Review Review
Takeoff and Climb Procedures Soft-Field Takeoff	 Introduction
Commercial Flight Maneuvers Chandelles Lazy Eights Steep Spirals	 Introduction Introduction Introduction
Ground Reference Maneuvers Eights on Pylons	 Introduction

Slow Flight Pitch/Airspeed Control Power/Altitude Control Climbs/Turns/Descents	 Review Review Review
Stalls Power-Off Stalls Power-On Stalls Accelerated Stalls	 Introduction Introduction Introduction
Approach and Landing Procedures Soft-Field Landing Go-Around/Rejected Landing	 Introduction Review
Postflight Operations Postflight Debrief	 Review
Special Emphasis Areas Checklist Usage Positive Aircraft Control	 Review Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this Lesson, the **PUI** will have gained an increased proficiency of basic aircraft avionics, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 20 degrees, altitude will be maintained within +/- 200 feet, and airspeed will be maintained within +/-10 kts during maneuvers. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 19-DL

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

During this lesson, the **IP** will continue training for emergency operations and commercial maneuvers during flight. The **PUI** will plan the most economical and efficient route, taking into consideration airspace, time, fuel, and distance to complete the lesson.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (As Applicable)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel or Castering Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Short-Field Takeoff	Review
Soft-Field Takeoff	Introduction

Emergency/Abnormal Procedures	
Engine Failure In Flight	Review
Systems and Equipment Malfunctions	Review
Commercial Flight Maneuvers	
Chandelles	Review
Lazy Eights	Review
Steep Spirals	Review
Ground Reference Maneuvers	
Eights on Pylons	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Climbs/Turns/Descents	Review
Stalls	
Power-Off Stalls	Review
Power-On Stalls	Review
Accelerated Stalls	Review
Approach and Landing Procedures	
Soft-field Landing	Review
180 Degree Power Off Accuracy Landing	Introduction
Go-Around/Rejected Landing	Review
Postflight Operations	
Securing Aircraft	Review
Postflight Inspection	Review
Postflight Debrief	Review
Special Emphasis Areas	
Positive Aircraft Control	Review
Collision Avoidance	Review
Runway Incursion Avoidance	Review
Aeronautical Decision Making	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained an increased proficiency of basic aircraft avionics, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 15 degrees, altitude will be maintained within +/- 150 feet, and airspeed will be maintained within +/- 10 kts during maneuvers. Additionally the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 20-DL

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

This lesson provides a review of navigation procedures, traffic pattern procedures, take-offs and landings.

For this lesson, the **PUI** will prepare a cross-country navigation log to demonstrate proper VFR navigation procedures to include E6B Usage, Pilotage, dead reckoning and diversion procedures.

All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures.

Lesson Content

SRM Risk Assessment Collision Avoidance Task Management Automation Management	 Review Review Review Review
Preflight Procedures Cockpit Management Preflight Briefing	 Review Review
Cockpit Familiarization Flight Controls Engine Controls	 Review Review
Checklist Usage Normal Procedures Checklist	 Review
Flows Cockpit Flows	 Review
Engine Start General Starting Procedures Engine Fire On Start Procedures	 Review Review
Before Takeoff Check Engine Check Instrument Check	 Review Review
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff	Review Review

Short-Field Takeoff Soft-Field Takeoff	Review
VFR Navigation Pilotage Dead Reckoning E6B Usage Diversion Procedures Open VFR Flight Plan (Simulate)	 Review Review Review Review Review
Approach and Landing Procedures Normal and/or Crosswind Landing Short-Field Landing Soft-Field Landing 180 Degree Power Off Accuracy Landing Go-Around/Rejected Landing	 Review Review Review Review Review
Postflight Operations Postflight Debrief	 Review
Special Emphasis Areas All Areas	 Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained an increased proficiency in cross country navigation procedures and traffic pattern operations. Additionally the **PUI** will exhibit increased SRM skills in all areas.

Landings will be made in accordance with ACS standards with the following modifications: on the 180 Degree Power Off Accuracy Landing- Airspeed -5/+10 knots, touchdown within -0/+300 feet of a specified point.

During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 21-DL

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

This Lesson provides a review of the procedures and skills needed in a single engine airplane.

For this lesson, the **PUI** will prepare a cross-country navigation log to demonstrate proper VFR navigation procedures to include E6B Usage, Pilotage, dead reckoning and diversion procedures.

All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Personal Weather Minimums Checklist	Review
Task Management	Review
Automation Management (As Applicable)	Review
Preflight Procedures	
Cockpit Management	Review
Preflight Inspection	Review
Preflight Briefing	Review
Route	
Most Economical Route Possible	Review
Most Efficient Route Possible	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review

Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	
Short-Field Takeoff	Review
Soft-Field Takeoff	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Steep Turns	
Steep Turns	Review
Commercial Flight Maneuvers	
Chandelles	Review
Lazy Eights	Review
Steep Spirals	Review
Ground Reference Maneuvers	
Eights on Pylons	Review
Stalls	
Power-Off Stalls	Review
Power-On Stalls	Review
Accelerated Stalls	Review
VFR Navigation	
Pilotage	Review
Dead Reckoning	Review
E6B Usage	Review
Diversion Procedures	Review
Open VFR Flight Plan (Simulate)	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Soft-Field Landing	Review
180 Degree Power Off Accuracy Landing	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Securing Aircraft	Review
Postflight Inspection	Review
Postflight Debrief	Review
Special Emphasis Areas	
All Areas	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson, the **PUI** will have gained an increased proficiency in cross country navigation procedures and traffic pattern operations. Additionally the **PUI** will exhibit increased SRM skills in all areas.

Landings will be made in accordance with Commercial ACS.

During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 22-GB

Equipment

Briefing Room

Lesson Objective

This lesson provides a review of pilot qualifications, certificates and documents, airworthiness requirements, weather information, VFR cross-country flight planning, national airspace system, performance, limitations, operations of systems, human factors/aeromedical factors, principles of pressurization, and supplemental oxygen.

Lesson Content

Oral	
Pilot Qualifications/Certificates and Documents	Review
Airworthiness Requirements	Review
Weather Information	Review
VFR Cross-Country Flight Planning	Review
National Airspace System	Review
Performance and Limitations	Review
Aerodynamics	Review
Operations of Systems	Review
Human Factors/Aeromedical Factors	Review
Principles of Pressurization	Review
Supplemental Oxygen	Review
Special Emphasis Areas	
All Areas	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson, the **PUI** must meet or exceed the outcome level specified with respect to the Commercial Pilot ACS.

Lesson 23-DL

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

This lesson provides a review of flight maneuvers and procedures in preparation for the End of Course flight lesson and the FAA check ride.

For this lesson, the **PUI** will prepare a cross-country navigation log to demonstrate proper VFR navigation procedures to include E6B Usage, Pilotage, dead reckoning and diversion procedures.

All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Personal Weather Minimums Checklist	Review
Task Management	Review
Automation Management (As Applicable)	Review
Preflight Procedures	
Preflight Inspection	Review
Cockpit Management	Review
Preflight Briefing	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Soft-Field Takeoff	Review

VFR Navigation	
Pilotage	Review
Dead Reckoning	Review
E6B Usage	Review
Diversion Procedures	Review
Open VFR Flight Plan (Simulate)	Review
Traffic Patterns	
Traffic Pattern Operations	Review
ATC Light Gun Signals	Review
Emergency/Abnormal Procedures	
Engine Failure In Flight	Review
Systems and Equipment Malfunctions	Review
Emergency Equipment and Survival Gear	Review
Steep Turns	
Steep Turns	Review
Stalls	
Power-Off Stalls	Review
Power-On Stalls	Review
Accelerated Stalls	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Commercial Flight Maneuvers	
Chandelles	Review
Lazy Eights	Review
Steep Spirals	Review
Ground Reference Maneuvers	
Eights on Pylons	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Soft Field Landing	Review
Short Field Landing	Review
Power-off 180º Accuracy Approach and Landing	Review
Emergency Approach and Landing	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Securing Aircraft	Review
Postflight Inspection	Review
Postflight Debrief	Review

Special Emphasis Areas All Areas	Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have met the approved FAA Airman Certification Standards for the Commercial Pilot Certificate with Airplane Single engine Class Rating privileges.

Lesson 24-STO–End of Course Stage Check (Oral)

Equipment

Briefing Area

Lesson Objective

The Chief/Assistant Chief or Designated End of Course Check Pilot will conduct the lesson. This lesson provides an evaluation of pilot qualifications/certificates and documents, airworthiness requirements, weather information, VFR cross-country flight planning, national airspace system, performance, limitations, operations of systems, human factors/aeromedical factors, principles of pressurization, and supplemental oxygen

Lesson Content

Oral Pilot Qualifications/Certificates and Documents Airworthiness Requirements Weather Information VFR Cross Country Flight Planning National Airspace System Performance and Limitations Aerodynamics Operations of Systems Human Factors/Aeromedical Factors Principles of Pressurization Supplemental Oxygen
Special Emphasis Areas All Areas
Additional Areas As Assigned By Instructor (If Necessary)

Completion Standards

At the completion of this lesson, the **PUI** will have knowledge of the tasks that meets or exceeds the outcome level specified with respect to the Commercial Pilot ACS.

Lesson 25-STF–End of Course Stage Check (Flight)

Equipment

SINGLE ENGINE AIRPLANE

Lesson Objective

The Chief/Assistant Chief or Designated End of Course Check Pilot will conduct the lesson. The **IP** will evaluate the **PUI** during all phases of the flight. All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance to complete the lesson.

Lesson Content

SRM Risk Assessment Collision Avoidance Personal Weather Minimums Checklist Task Management Automation Management (As Applicable) Flight Deck Management
Preflight Procedures Lesson Planning Preflight Inspection Preflight Briefing
Checklist Usage Normal Procedures Checklist Abnormal Procedures Checklist Emergency Procedures Checklist
Flows Cockpit Flows
Engine Start General Starting Procedures
Before Takeoff Check Engine Check Instrument Check
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff Short-Field Takeoff Soft-Field Takeoff
Traffic Patterns Traffic Pattern Operations ATC Light Gun Signals

Emergency/Abnormal Procedures Engine Failure In Flight Systems and Equipment Malfunctions Emergency Equipment and Survival Gear
Slow Flight Pitch/Airspeed Control Power/Altitude Control
Steep Turns Steep Turns
Commercial Flight Maneuvers Chandelles Lazy Eights Steep Spirals
Ground Reference Maneuvers Eights on Pylons
Stalls Power-Off Stalls Power-On Stalls Accelerated Stalls
VFR Navigation Pilotage Dead Reckoning Navigation Systems and Radar Services Diversion Procedures Open VFR Flight Plan (Simulate)
Approach and Landing Procedures Normal and/or Crosswind Landing Short-field Landing Soft-Field Landing Emergency Approach and Landing Power-Off 180º Accuracy Approach and Landing Go-Around/Rejected Landing
Postflight Operations Securing Aircraft Postflight Inspection Postflight Debrief
Special Emphasis Areas All Areas

Additional Areas As Assigned By Instructor (If Necessary)

Completion Standards

At the completion of this lesson the **PUI** will have exceeded the approved Airman Certification Standards for the Commercial Pilot Certificate with Airplane Single engine Class Rating privileges. Additionally, they student will have met all Aeronautical Experience requirements as required by part 141 and outlined in this chapter.